

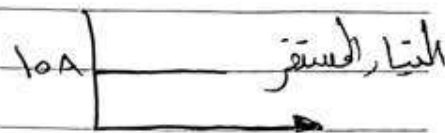
Subject: Lec - 1 -

Date: 14/11/2023 الموافق

مكتبة جامعة الزيتونة
الكلية الهندسية الكهربائية
التل 07707851221

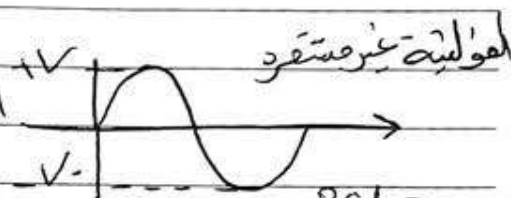
Fundamentals of Electrical Engineering

Dc : Direct current



لا يتغير مقدار
التيار مع الزمن

Ac : Alternative current



220V
يمكن نقله بالأسلاك
الكهربائية

Bridge Dc
220Ac → 220Dc

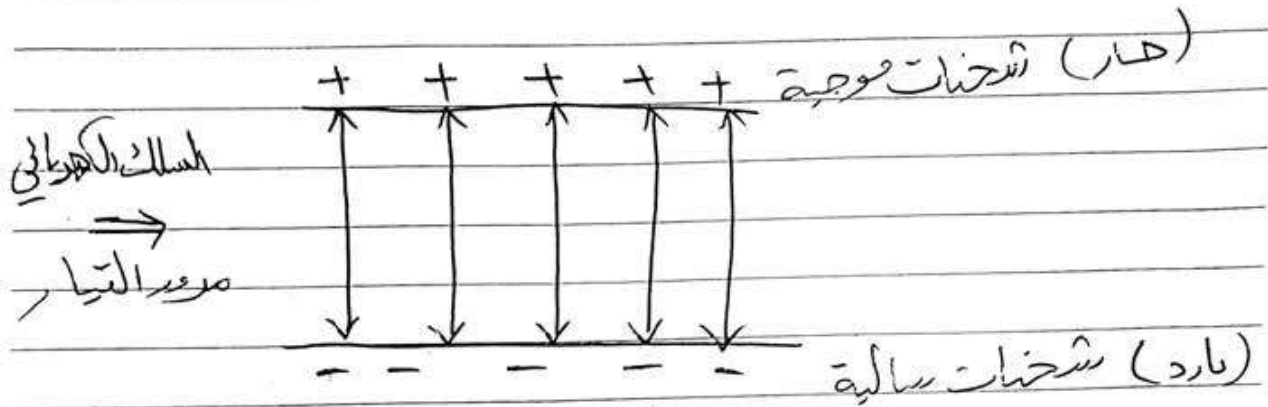
Voltage [V] : The potential difference

between two points. فرق الجهد بين نقطتين

Current [I] : The rate of charge moving in a certain metal.

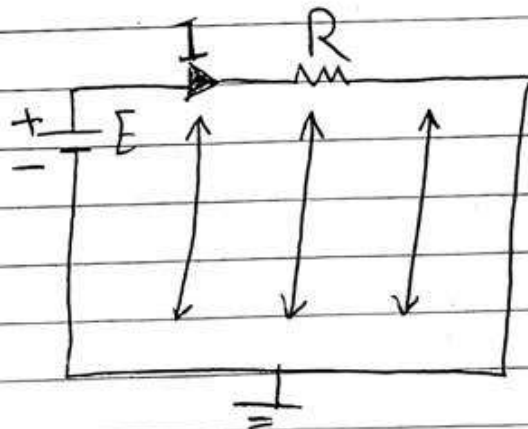
عدد الشحنات التي
تتحرك خلال سلك معين

Assistant Lecturer : — Doaa Yousif



فائدة الشحنات السالبة ليست نقل الفولتية لـ 8 الفولتية هي
فوق الجهد بين نقطتين

Voltage Source مصدر الفولتية



$R =$ مقاومة لتقاوم مرور التيار

تتبع مرور التيار دون عائق

لذلك كلما تزداد قيمة R يقل

التيار. وكل جهاز كهربائي

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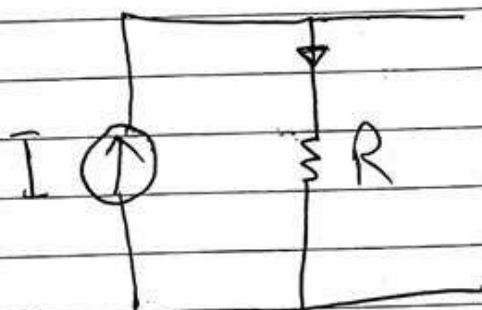
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هو مقاومة حسب مقارنته للتيار، يحدد كم قيمة التيار الذي يمر.

Current source

مصدر التيار



لإخترافي واستخدمه قليل

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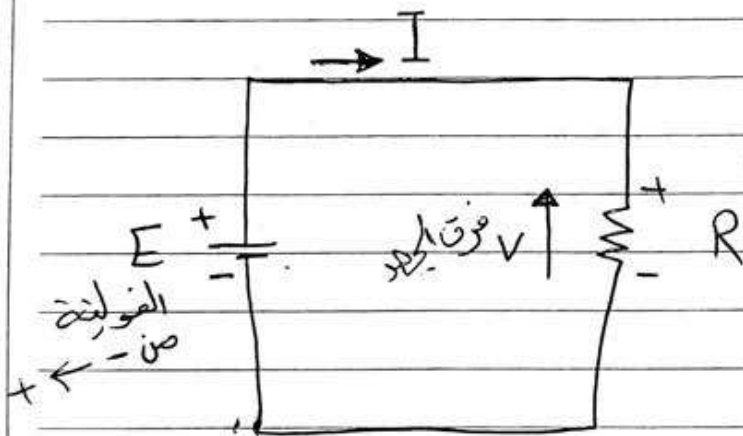
/ / التاريخ:

Ohm's Law

$$\checkmark R = \frac{V}{I}, I = \frac{V}{R}, V = I \cdot R$$

$$R [\Omega], V [V], I [A]$$

$\downarrow$ ohm $\downarrow$ volt $\downarrow$ Ampere



مقاومة صيغته توافقي
الخلية متساوية
 $V = E$

Power القدرة

$$\checkmark P = V \cdot I, R = \frac{V}{I}$$

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$$P = R \cdot I \cdot I$$

$$\checkmark P = I^2 \cdot R$$

$$\checkmark P = \frac{V^2}{R}$$

Conductance G الموصلية

$$\checkmark G = \frac{1}{R}$$

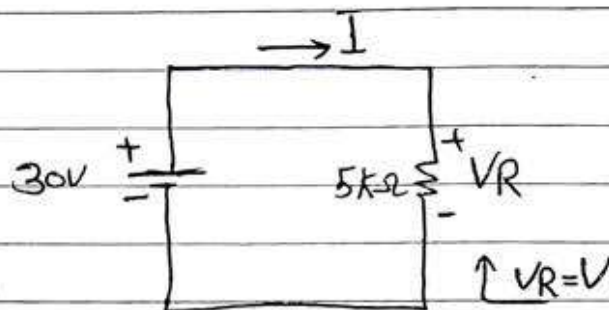
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Example 8 $\Rightarrow$ In the circuit shown, Calculate the current I , the conductance G , and the power P



Solution: -

$$I = \frac{V}{R} = \frac{30}{5 \times 10^3} = 6 \times 10^{-3} = 6 \text{ mA}$$

$$G = \frac{1}{R} = \frac{1}{5 \times 10^3} = 0.2 \times 10^{-3} = 0.2 \text{ mS}$$

$$P = V \cdot I = 30(6 \times 10^{-3}) = 180 \text{ mW}$$

or

$$P = I^2 R = (6 \times 10^{-3})^2 (5 \times 10^3) = 180 \text{ mW}$$

or

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~~P.V. =~~

Ex: — Find the current through a resistor of resistance $R = 2\ \Omega$ if the voltage across the resistor is $6V$.

Sol: Substitute R by 2 and V by 6 in Ohm's Law

$$V = RI$$

$$6 = 2 \times I$$

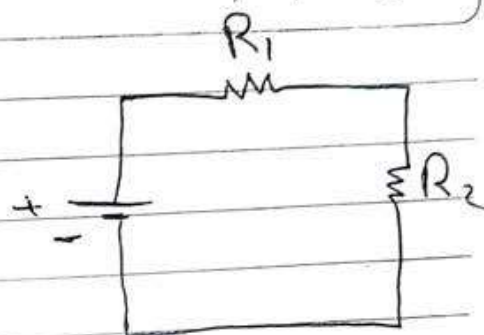
$$I = \frac{6}{2} = 3A$$

Ex: In the circuit below resistors R_1 and R_2 are in series and have resistances of $5\ \Omega$ and $1\ \Omega$ respectively. The voltage across resistor R_1 is $4V$. Find the current passing through resistor

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and the voltage across
the same resistor



Soln $V_1 = R_1 I_1$

$$4 = 5 \times I_1 \Rightarrow I_1 = \frac{4}{5} = 0.8 \text{ A}$$

The two Resistor is series $I_1 = I_2 = 0.8 \text{ A}$

$$V_2 = R_2 I_2 = 10 \times 0.8 \text{ A} = 8 \text{ V}$$

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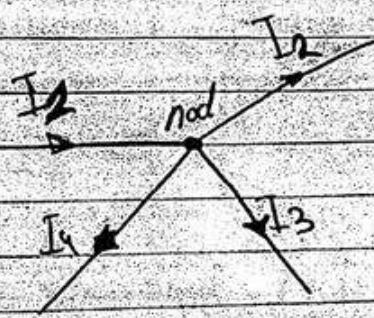
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Kirchoff's Laws

مجموعة من النقاط
مع الدوائر الكهربائية

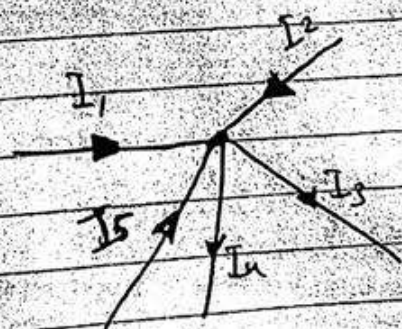
1) Kirchoff's current Law [KCL]



nod = نقطة التقاء
التيار إلى عدة خطوط (الفرع)

وهذه التفرعات يمكن تكون
متسلسلة أو غير متسلسلة
حسب قيمة R

$$I_1 = I_2 + I_3 + I_4$$

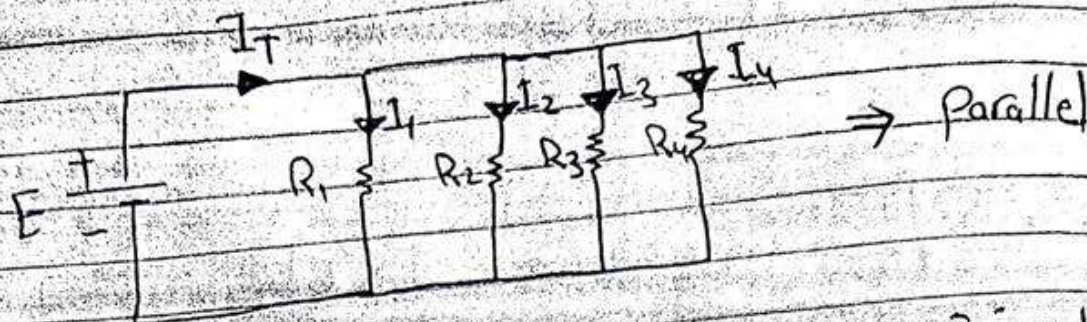


$$I_1 + I_5 = I_2 + I_3 + I_4$$

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التيارات تجمع في قيمة المقادير وهي تختلف باختلاف قيمة المقادير ولكن مجموع التيارات الخارجة = الداخلة

$$I_T = I_1 + I_2 + I_3 + I_4$$

مجموع التيارات الخارجة = التيارات الداخلة

∴ parallel ∴ $E = V_{R1} = V_{R2} = V_{R3} = V_{R4}$

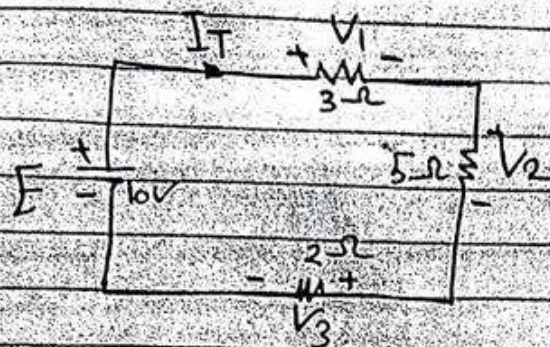
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Ex) For the circuit shown:

find the voltage V_1, V_3 .

Sol) Ohm's Law $I_T = \frac{V_T}{R_T}$

$$R_T = R_1 + R_2 + R_3$$

$$I_T = \frac{10V}{10\Omega}$$

$$= 3 + 5 + 2$$

$$I_T = 1A$$

$$= 10\Omega$$

Series $\therefore I_T = I_{R_1} = I_{R_2} = I_{R_3}$

$$V_1 = I_{R_1} \cdot R_1$$

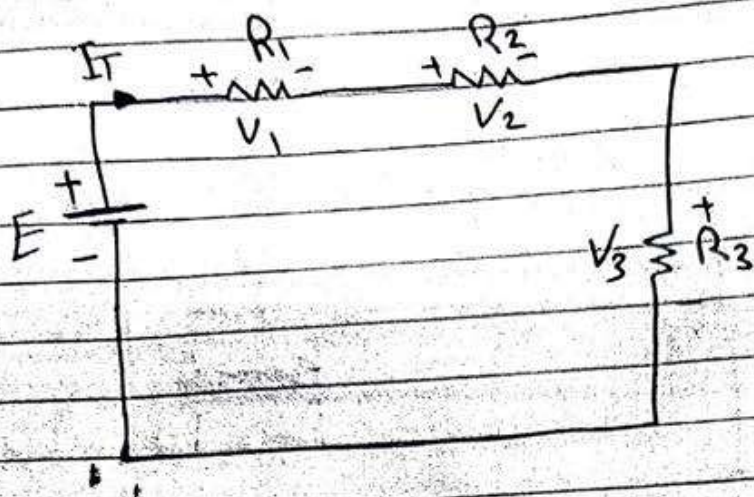
$$V_3 = I_{R_3} \cdot R_3$$

$$= 1 \cdot 3 = 3V$$

$$= 1 \cdot 2 = 2V$$

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* Series $\Rightarrow R_T = R_1 + R_2 + R_3$

$$E = V_1 + V_2 + V_3$$

$$E - V_1 - V_2 - V_3 = 0$$

$$I_T = I_{R_1} = I_{R_2} = I_{R_3}$$

Subject:

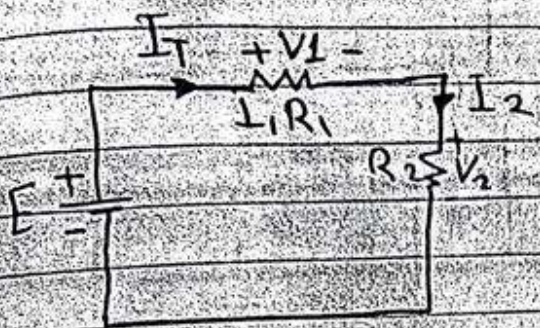
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2) Kirchhoff's Voltage Law [KVL]



→ Series

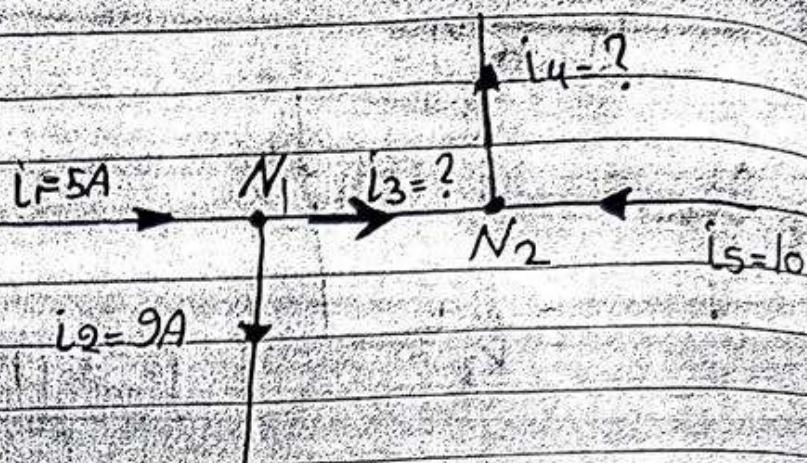
$$I_T = I_{R1} = I_{R2}$$

$$E = V_1 + V_2$$

$$E - V_1 - V_2 = 0$$

[مجموع الفولتيات = الفولتية الكلية] ⇒ KVL
لحل المقاومات

Ex) Find currents i_3 and i_4 at the nodes N_1 and N_2 shown below.



Sol) At node N_1 , i_1 flows into N_1 and i_2 and i_3 flow out N_1

$$i_1 = i_3 + i_2$$

$$5 = i_3 + 9$$

$$i_3 = -4 \text{ into node } N_1$$

At node N_2 , i_3 and i_5 flow and i_4 flows out of N_2 ,

$$i_3 + i_5 = i_4$$

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$$-4 + 10 = I_4$$

$$I_4 = 6 \quad \text{out of node } N_2$$

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Voltage Divider Rule

VDR

قانون مجزء الفولتية يستخدم عندما يكون السرب series

 V_T

$$V_X = \frac{E \cdot R_x}{R_T}$$

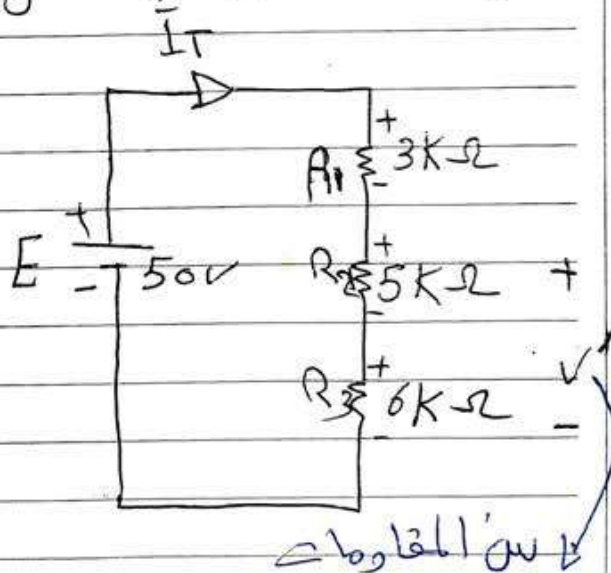
اي نقطة

Ex) Determine the voltage V_1 , V_2 and V' for the circuit shown:

Sol, ∴ Series

$$\therefore R_T = R_1 + R_2 + R_3$$

$$= 3 + 5 + 6 = 14 \text{ K}\Omega$$



$$V_1 = \frac{50 \times 3}{14} = \frac{150}{14} = 10.7 \text{ V}$$

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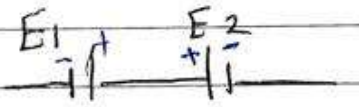
$$V_2 = \frac{E \cdot R_2}{R_T} = \frac{50 \times 5}{14} = \frac{250}{14} = 17.85V$$

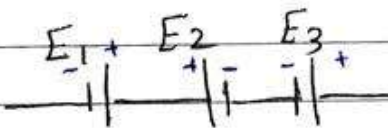
$$V' = \frac{E \cdot R'}{R_T} \quad , \quad R' = R_2 + R_3$$

$$= 5 + 6 = 11K\Omega$$

$$V' = \frac{50 \times 11}{14} = \frac{550}{14} = 39.28V$$

Notes: - ①  $E_T = E_1 + E_2$

②  $E_T = E_1 - E_2$

③  $E_T = E_1 + E_3 - E_2$

$$R' = R_x + R_x \quad \text{توالي}$$

$$R' = \frac{R_x \times R_x}{R_x + R_x} \quad \text{توازي}$$

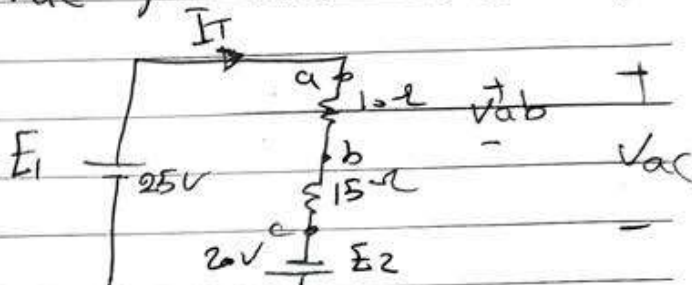
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Ex) find V_{ab} and V_{ac} for the cct shown

$$\text{Sol)} \quad E_T = E_1 + E_2 \Rightarrow \text{---|---|---}$$

$$= 25 + 20$$

$$= 45V$$

$$R_T = 10 + 15$$

$$= 25\Omega$$

$$V_{ab} = \frac{E_T \cdot R_{ab}}{R_T} = \frac{45 \cdot 10}{25} = 18V$$

$$\Rightarrow R_{ac} = R_1 + R_2 = 10 + 15 = 25\Omega$$

$$V_{ac} = \frac{E_T \cdot R_{ac}}{R_T} = \frac{45 \cdot 25}{25} = 45V$$

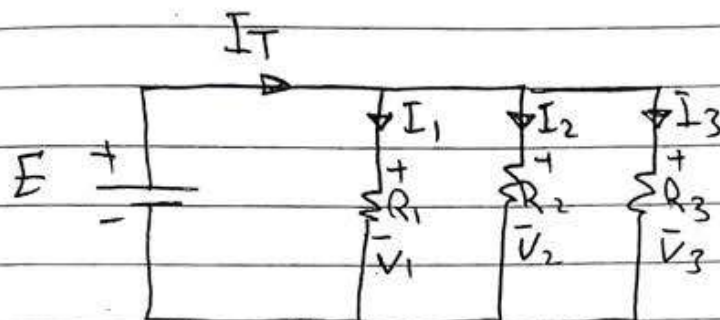
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Parallel circuit



Parallel

$$\textcircled{1} \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\textcircled{2} \quad I_1 = \frac{V_1}{R_1}, \quad I_2 = \frac{V_2}{R_2}, \quad I_3 = \frac{V_3}{R_3}$$

$$\textcircled{3} \quad V_1 = V_2 = V_3 = E$$

لكل الفرع الرئيسية الرئيسية غير صفرية سوف

نتيجة القانون Current divider Rule

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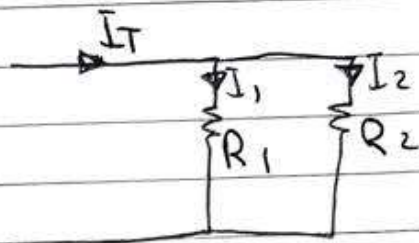
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Current divider Rule

$$I_1 = \frac{I_T \cdot R_2}{R_1 + R_2}$$

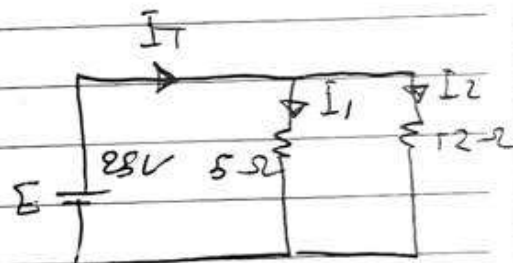


لا يُستخدم قانون Ohm
لعدم وجود E

$$I_2 = \frac{I_T \cdot R_1}{R_1 + R_2}$$

Ex) for the cct shown

① determine I_T and I_1 and I_2



② determine the power to each resistor

$$\text{So) } \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{5} + \frac{1}{12} = \frac{17}{60} = 3.5 \text{ S}$$

$$I_T = \frac{E}{R_T} = \frac{25}{3.5} = 7.1 \text{ A}$$

$$I_1 = \frac{V_1}{R_1} = \frac{25}{5} \Rightarrow E = V_1 = V_2$$

DELUXE

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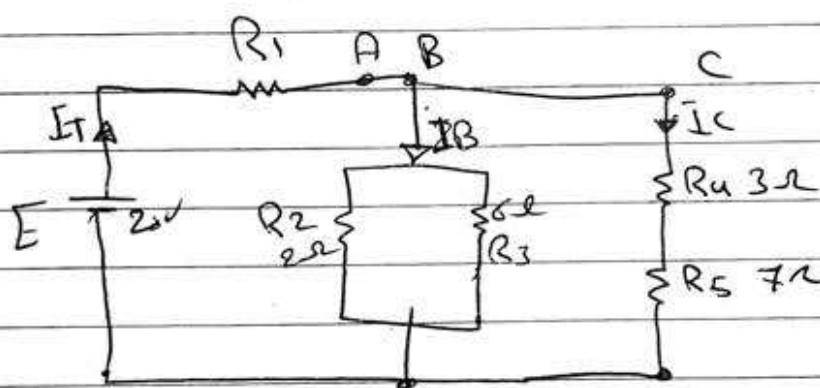
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$$I_2 = \frac{V_2}{R_2} = \frac{25}{12} = 2.1A$$

$$P_1 = V_1 \cdot I_1 = 25 \cdot 5 = 125 \text{ wat}$$

$$P_2 = V_2 \cdot I_2 = 25 \cdot 2.1 = 52.5W$$

Ex) For circuit shown, find V_A, V_B, V_C



Sol) $R_C = R_5 + R_6 = 3 + 7 = 10 \Omega$ Series

parallel $R_B \Rightarrow \frac{1}{R_B} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{2} + \frac{1}{6}$ توازي

$$R_B = 1.5 \Omega$$

$$R_T = R_1 + R_B + R_C + R_4 + R_6$$

$$R_T =$$

$$V_C = \frac{E \cdot R_C}{R_T}$$

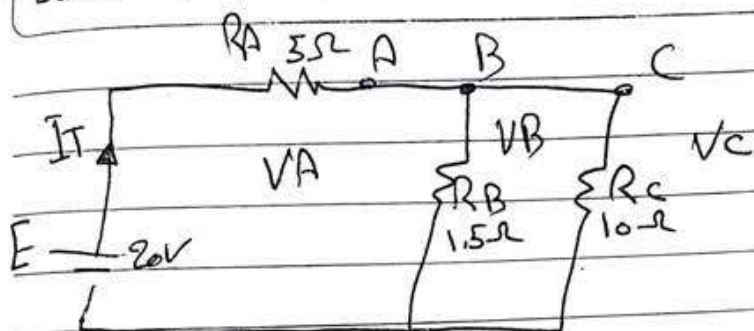
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$$V_A = I_A \cdot R_A$$

$$R_T = R_A + (R_B // R_C)$$

$$R_T = 5 + \frac{R_B \cdot R_C}{R_B + R_C}$$

$$R_T = R_A + 1.3$$

$$R_T = 5 + 1.3$$

$$R_T = 6.3 \Omega$$

$$I_T = \frac{E}{R_T} = \frac{20}{6.3} = 3.174 \text{ A}$$

$$V_A = I_A \cdot R_A \rightarrow I_T = I_A$$

$$V_A = 3.174 \cdot 5 = 15.87 \text{ V}$$

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According to CDR

$$I_B = \frac{I_T \cdot R_C}{R_B + R_C} = \frac{3.174 \times 10}{10 + 1.5} = 2.76A$$

$$I_C = \frac{I_T \cdot R_B}{R_B + R_C} = \frac{3.174 \times 1.5}{1.5 + 10} = 0.414A$$

$$V_B = I_B \cdot R_B \Rightarrow 2.76 \times 1.5 = 4.14V$$

$$V_C = I_C \cdot R_C \Rightarrow 0.414 \times 10 = 4.14V$$

because its parallel

$$\therefore V_B = V_C$$

Lec-4-

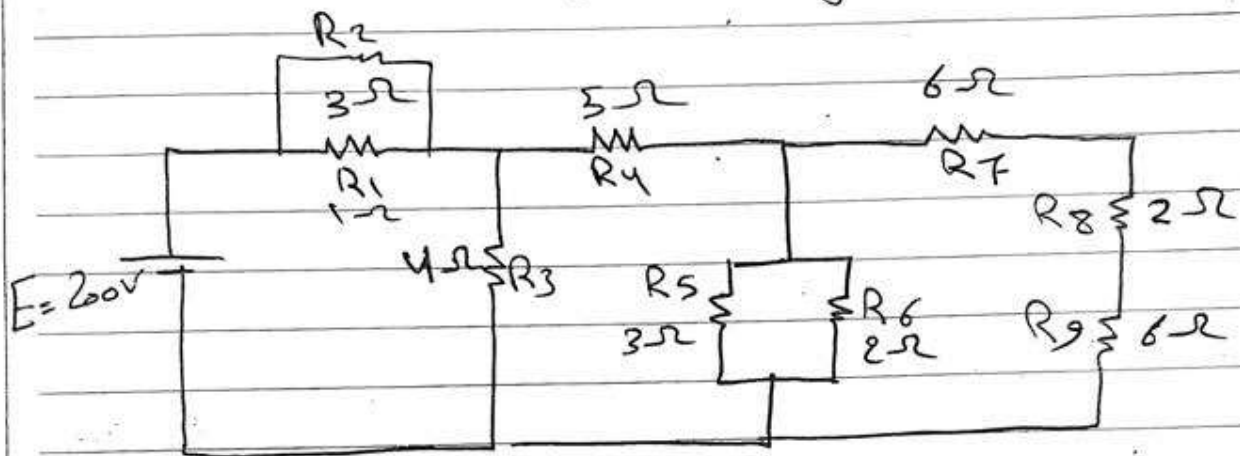
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Ex) For the network shown, determine all currents and voltages through all branches.

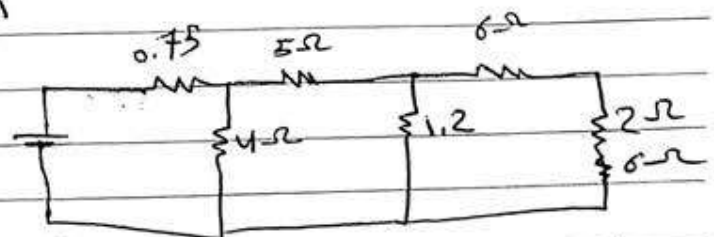


$$R_{12} = R_1 // R_2 \Rightarrow R_{12} = \frac{R_1 * R_2}{R_1 + R_2}$$

$$R_{12} = \frac{3 * 1}{3 + 1} = \frac{3}{4} = 0.75 \Omega$$

$$R_{56} = R_5 // R_6 \Rightarrow R_{56} = \frac{R_5 * R_6}{R_5 + R_6}$$

$$R_{56} = \frac{3 * 2}{3 + 2} = \frac{6}{5} = 1.2 \Omega$$



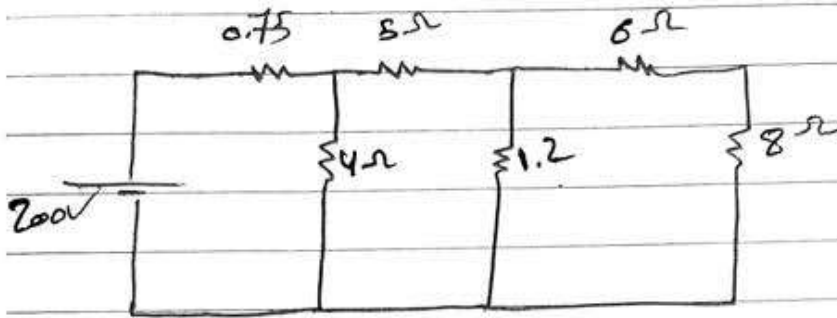
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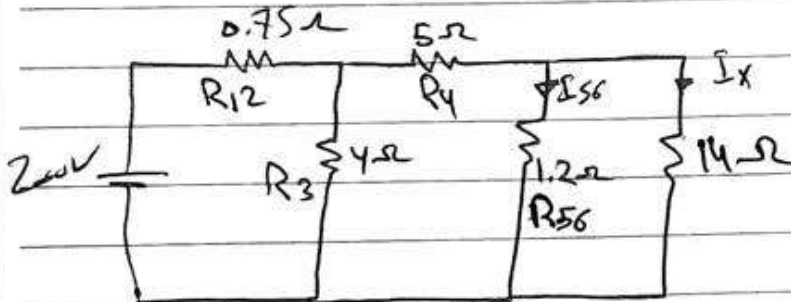
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$$R_{89} = R_8 + R_9 = 2 + 6 = 8\Omega$$

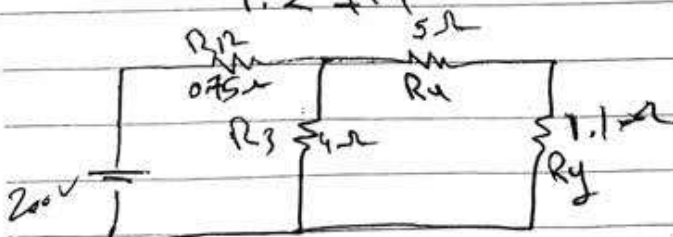


$$R_x = R_7 + R_8 + R_9 = 6 + 2 + 6 = 14\Omega$$



$$R_y = R_x // R_{56} = \frac{R_x * R_{56}}{R_x + R_{56}}$$

$$R_y = \frac{1.2 * 14}{1.2 + 14} = 1.1\Omega$$



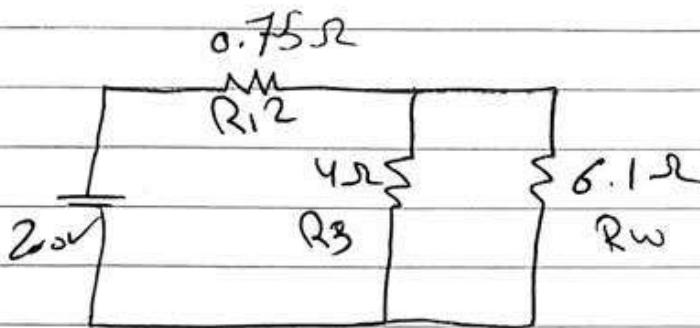
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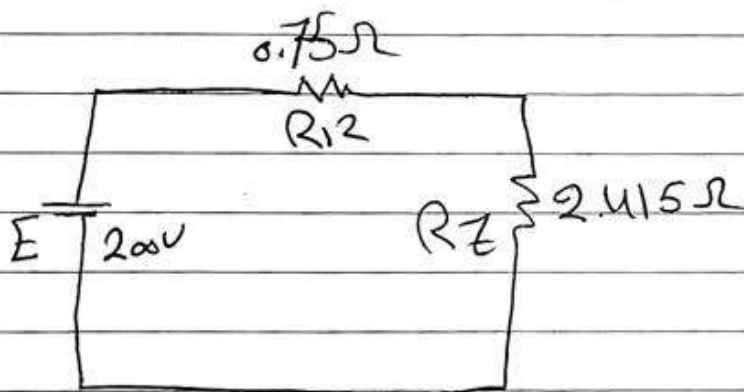
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$$R_w = R_y + R_u = 5 + 1.1 = 6.1 \Omega$$



$$R_Z = R_3 // R_w = \frac{R_3 * R_w}{R_3 + R_w} = \frac{4 * 6.1}{4 + 6.1} = 2.415 \Omega$$



$$R_T = R_{12} + R_Z$$

$$R_T = 0.75 + 2.415$$

$$R_T = 3.165 \Omega$$

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$$I_T = \frac{E}{R_T} = \frac{200}{3.165} = 63.19 A$$

$$I_T = I_{12} = 63.19 A$$

$$V_{12} = V_1 = V_2$$

$$V_{12} = I_T \times R_{12} \Rightarrow 63.19 \times 0.75$$

$$V_{12} = 47.4 V = V_1 = V_2 \quad R_1 // R_2$$

$$I_1 = \frac{V_1}{R_1} = \frac{47.4}{1} = 47.4 A$$

$$I_2 = \frac{V_2}{R_2} = \frac{47.4}{3} = 15.8 A$$

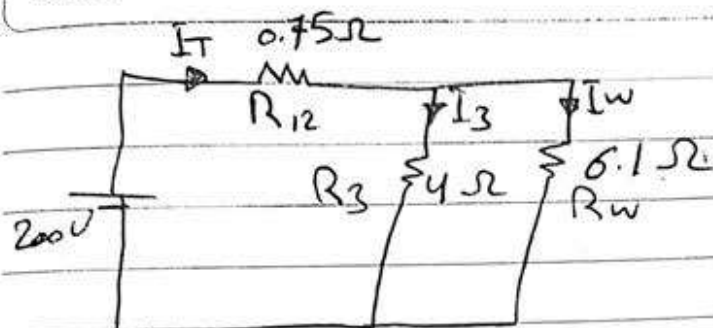
or using CDR

$$I_1 = \frac{I_T \times R_2}{R_1 + R_2} \Rightarrow \frac{63.19 \times 3}{3 + 1} = 47.4 A$$

$$I_2 = \frac{I_T \times R_1}{R_1 + R_2} \Rightarrow \frac{63.19 \times 1}{1 + 3} = 15.8 A$$

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Using CDR

$$I_3 = \frac{I_{12} * R_W}{R_3 + R_W} = \frac{63.19 * 6.1}{4 + 6.1} = 38.164 A$$

$$V_3 = I_3 * R_3 = 152.6 V$$

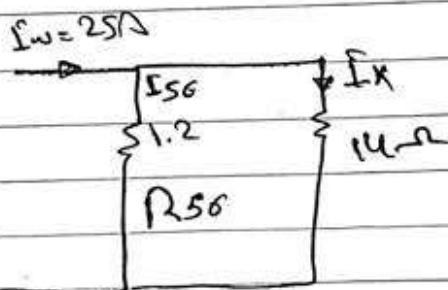
$$I_W = 25 A$$

$$R_W = R_Y + R_U$$

$$R_Y = R_X // R_{56}$$

$$I_U = I_W = 25 A$$

$$V_U = I_U * R_U = 25 * 5 = 125 V$$



Subject:

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$$I_{56} = \frac{I_w \times R_x}{R_{56} + R_x} = \frac{25 \times 14}{24 + 1.2} = 23A$$

$$I_x = I_w - I_{56} = 2A$$

$$R_x = R_7 + R_8 + R_9 \quad \text{Series}$$

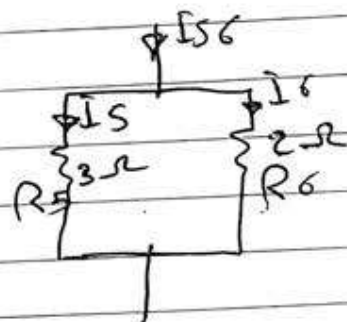
$$I_x = I_7 = I_8 = I_9 = 2A$$

$$V_7 = I_7 \times R_7 = 2 \times 6 = 12V$$

$$V_8 = I_8 \times R_8 = 2 \times 2 = 4V$$

$$V_9 = I_9 \times R_9 = 2 \times 6 = 12V$$

$$R_{56} = R_5 \parallel R_6$$



Subject:

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$$I_5 = \frac{I_{56} * R_6}{R_5 + R_6} = 9.2A$$

$$I_6 = 23 - 9.2 = 13.8A$$

$$V_5 = V_6 = I_5 * R_5 \Rightarrow 9.2 * 3 = 27.6V$$

Subject:

Lec - 5 -

Date:

الدوام:

موضوع الدرس:

التاريخ:

Loop Current Method [Mesh]

Determinants

هناك نوع من الدوائر الكهربائية لا يطبق
عليها توافقي ولا التوالي وانما تستخدم
فيها وتحل باستخدام $K1$

$$a_1 x + b_1 y = c_1$$

$$a_2 x + b_2 y = c_2$$

$$x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1}$$

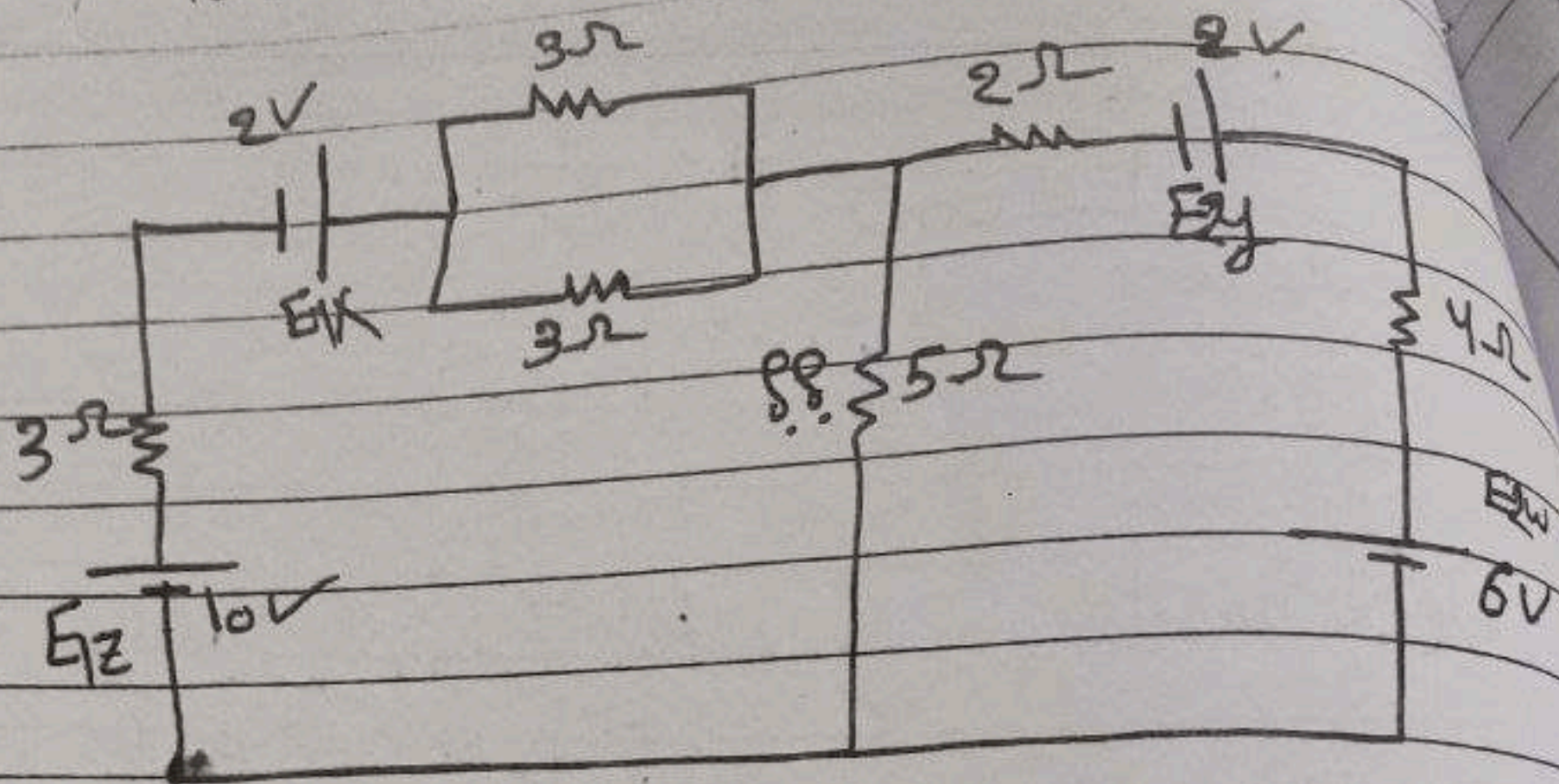
$$y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1}$$

Subject:

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الموافق:

Ex:-



At Loop (1)

$$E_2 + E_1 = E_1 \rightarrow E_1 = 10 + 2 = 12V$$

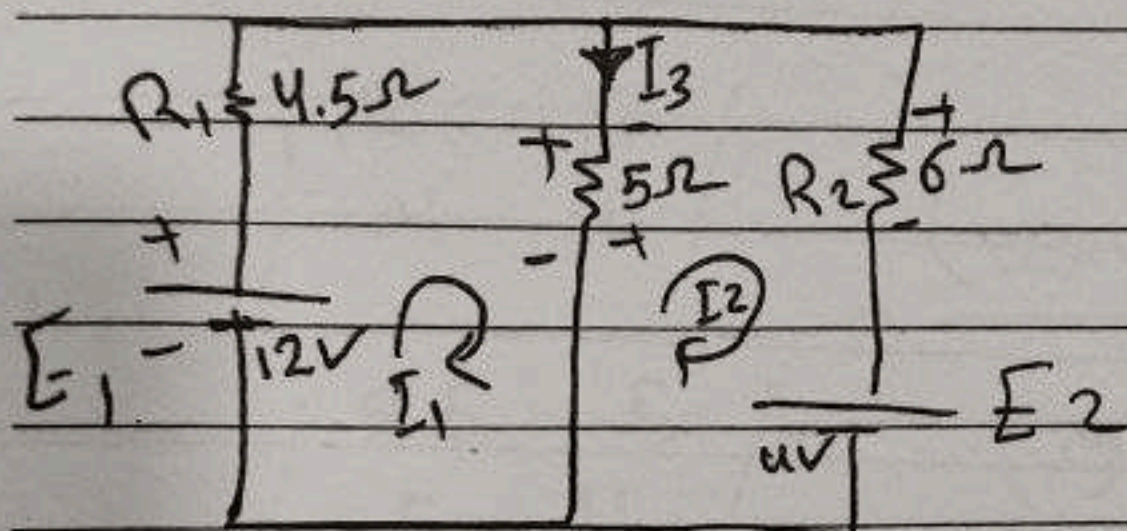
$$R_x = 3 \parallel 3 = \frac{3 * 3}{3 + 3} = 1.5 \Omega$$

$$R_1 = 3 \Omega + R_x = 3 + 1.5 = 4.5 \Omega$$

At loop (2)

$$E_2 = E_{2W} - E_{2y} \Rightarrow 6 - 2 = 4V$$

$$R_2 = 2 + 4 = 6\Omega$$



Method (1)

$$\text{Loop (1)} \rightarrow E_1 = V_1 + V_3$$

$$I_1 - V_1 - V_3 = 0$$

$$12 - I_1 R_1 - (I_1 - I_2) R_3 = 0$$

$$12 - 4.5 I_1 - 5(I_1 - I_2) = 0$$

$$12 - 4.5 I_1 - 5 I_1 + 5 I_2 = 0$$

Subject:

Date: / /

الموافق:

$$12 - 9.5I_1 + 5I_2 = 0$$

$$12 - 9.5I_1 + 5I_2 \quad \text{---} \quad \textcircled{1}$$

Loop (2)

$$-E_2 - V_3 - V_2 = 0$$

$$-4 - R_3(I_2 - I_1) - R_2 I_2 = 0$$

$$-4 - 5(I_2 - I_1) - 6I_2 = 0$$

$$-4 - 5I_2 + 5I_1 - 6I_2 = 0$$

$$-4 - 11I_2 + 5I_1 = 0$$

$$-4 = -5I_1 + 11I_2 \quad \text{---} \quad \textcircled{2}$$

$$12 = 9.5 I_1 - 5 I_2 \quad \text{--- ①}$$

$$-4 = -5 I_1 + 11 I_2 \quad \text{--- ②}$$

Solving by determinants:

$$I_1 = \frac{\begin{vmatrix} 12 & -5 \\ -4 & 11 \end{vmatrix}}{\begin{vmatrix} 9.5 & -5 \\ -5 & 11 \end{vmatrix}}} = \frac{(12 \times 11) - (-5 \times -4)}{\begin{vmatrix} 9.5 & -5 \\ -5 & 11 \end{vmatrix}}}$$

$$= \frac{112}{79.5}$$

$$\therefore I_1 = 1.4 A$$

$$I_2 = \frac{\begin{vmatrix} 9.5 & 12 \\ -5 & -4 \end{vmatrix}}{\begin{vmatrix} 9.5 & -5 \\ -5 & 11 \end{vmatrix}}} = \frac{(9.5 \times -4) - (-5 \times 12)}{79.5} = \frac{22}{79.5}$$

$$\therefore I_2 = 0.27 A$$

Subject:

Date: / /

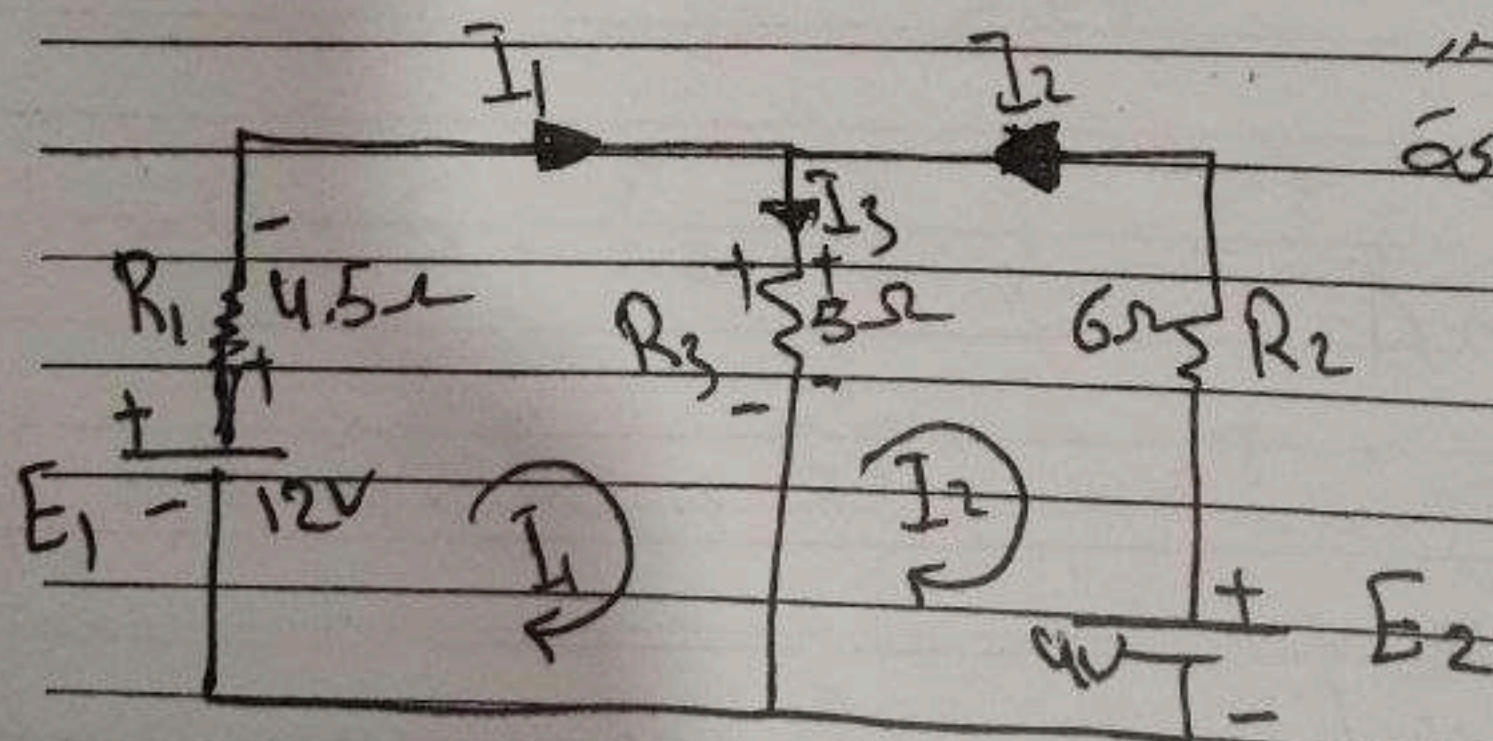
الموافق:

$$\therefore I \text{ at } 5\Omega \text{ or } R \quad I_3 = I_1 - I_2$$

$$\therefore I_3 = 1.4 - 0.27$$

$$I_3 = 1.13A$$

Method (2)



يتم التيار في اتجاه اليمين
ليس بالحدس قارب الساتر

Loop (1)

$$E_1 - V_1 - V_3 = 0 \Rightarrow 12 - R_1 I_1 - R_3 (I_1 + I_2) = 0$$

$$12 - 4.5 I_1 - 5 (I_1 + I_2) = 0$$

$$12 - 4.5 I_1 - 5 I_1 - 5 I_2 = 0$$

$$12 - 9.5 I_1 - 5 I_2 = 0$$

$$12 = 9.5 I_1 + 5 I_2 \quad \text{--- (1)}$$

at Loop (2)

$$E_2 - V_2 - V_3 = 0$$

$$4 - 6 I_2 - 5 (I_1 + I_2) = 0$$

$$4 - 11 I_2 - 5 I_1 = 0$$

$$4 = 5 I_1 + 11 I_2 \quad \text{--- (2)}$$

Subject:

الموافق:

Date:

$$12 = 9.5 I_1 + 5 I_2 \quad \text{--- (1)}$$

$$4 = 5 I_1 + 11 I_2 \quad \text{--- (2)}$$

$$I_1 = \frac{\begin{vmatrix} 12 & 5 \\ 4 & 11 \end{vmatrix}}{\begin{vmatrix} 9.5 & 5 \\ 5 & 11 \end{vmatrix}} = \frac{(12 \times 11) - (4 \times 5)}{(9.5 \times 11) - (5 \times 5)} = \frac{112}{79.5}$$

$$I_1 = 1.4 A$$

$$I_2 = \frac{\begin{vmatrix} 9.5 & 12 \\ 5 & 4 \end{vmatrix}}{79.5} = \frac{9.5 \times 4 - 5 \times 12}{79.5}$$

$$= \frac{-22}{79.5}$$

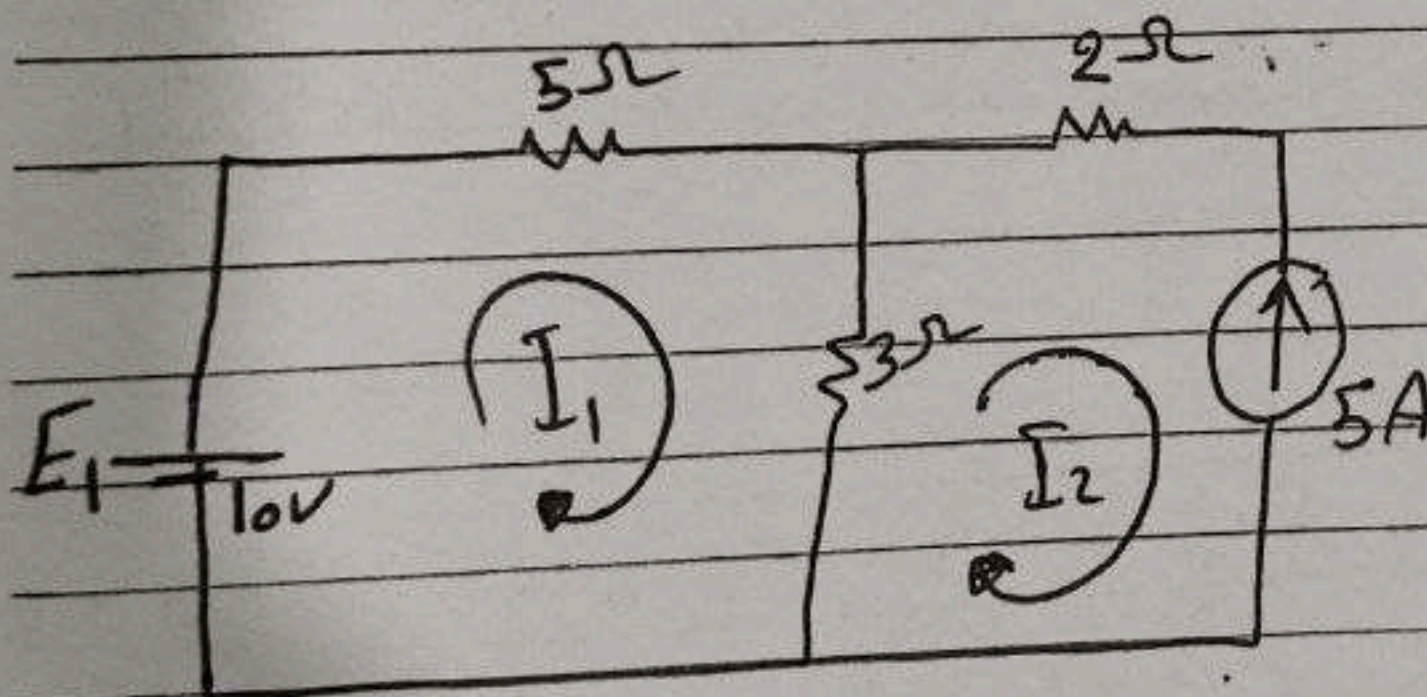
$$= -0.27 A$$

$$I \text{ at } 5\Omega \text{ or } I_3 = I_1 + I_2$$

$$I_3 = 1.4 + (-0.27)$$

$$I_3 = 1.13A$$

Ex) Find I_3



at Loop(1)

$$E - V_1 - V_3 = 0 \Rightarrow 10 - 5I_1 - 3(I_1 - I_2) = 0$$

$$10 - 5I_1 - 3I_1 + 3I_2 = 0$$

$$10 - 8I_1 + 3I_2 = 0$$

$$I_2 \text{ is known, } I_2 = 5A$$

$$\therefore 10 - 8I_1 + 3 \times 5 = 0$$

$$10 - 8I_1 - 15 = 0$$

$$8I_1 = -5 \rightarrow I_1 = -0.625A$$

$$\text{At } I_3 = I_1 - I_2$$

$$= -0.625 - 5$$

$$= -5.625A$$

Subject: _____

Lec. 6 : موضوع الدرس

Date : / / : التاريخ

/ / : التاريخ

Determinants

$$a_1 x + b_1 y + c_1 z = d_1$$

$$a_2 x + b_2 y + c_2 z = d_2$$

$$a_3 x + b_3 y + c_3 z = d_3$$

$$x = \frac{\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$

$$y = \frac{\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$

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موضوع الدرس: _____

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التاريخ: ____ / ____ / ____

$$Z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} \begin{vmatrix} c_1 \\ c_2 \\ c_3 \end{vmatrix}$$

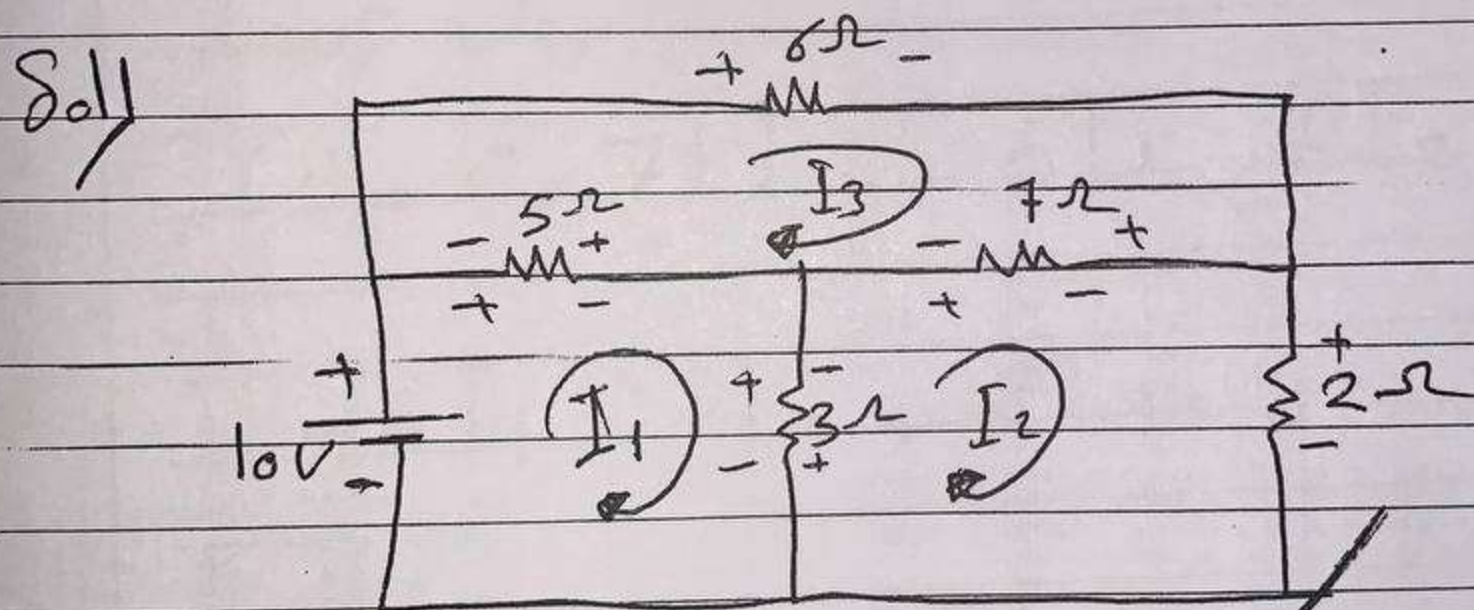
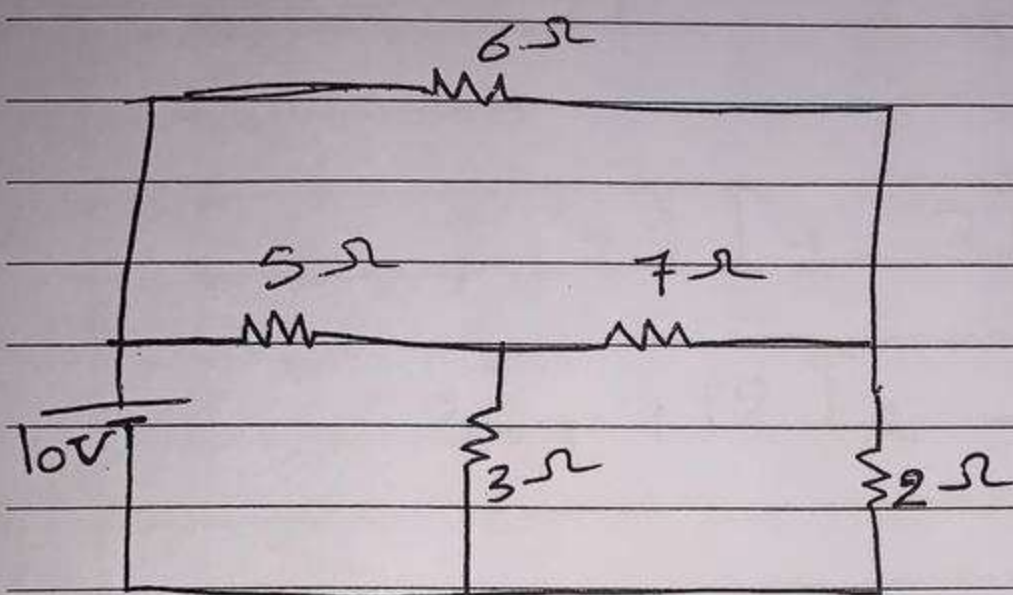
$$\therefore a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3 - [a_3 b_2 c_1 + b_3 c_2 a_1 + c_3 a_2 b_1]$$

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Ex: Find the current at 5Ω 

Loop (1)

$$(5 + 3)I_1 - 3I_2 - 5I_3 = 10$$

$$8I_1 - 3I_2 - 5I_3 = 10 \quad \text{--- (1)}$$

Loop (2)

$$(3 + 7 + 2) I_2 - 3I_1 - 7I_3 = 0$$

$$-3I_1 + 12I_2 - 7I_3 = 0 \quad \text{--- (2)}$$

Loop (3)

$$(5 + 6 + 7) I_3 - 5I_1 - 7I_2 = 0$$

$$-5I_1 - 7I_2 + 18I_3 = 0 \quad \text{--- (3)}$$

$$8I_1 - 3I_2 - 5I_3 = 10$$

$$-3I_1 + 12I_2 - 7I_3 = 0$$

$$-5I_1 - 7I_2 + 18I_3 = 0$$

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$$I_1 = \frac{\begin{vmatrix} 10 & -3 & -5 & | & 10 & -3 \\ 0 & 12 & -7 & | & 0 & 12 \\ 0 & -7 & 18 & | & 0 & -7 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 & | & a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 & | & a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 & | & a_3 & b_3 & c_3 \end{vmatrix}}$$

$$\begin{vmatrix} 8 & -3 & -5 & | & 8 & -3 \\ -3 & 12 & -7 & | & -3 & 12 \\ -5 & -7 & 18 & | & -5 & -7 \end{vmatrix}$$

$$I_1 = \frac{\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$

$$\Delta = 8 \times 12 \times 18 + (-3 \times -7 \times -5) + (-5 \times -3 \times -7) - [(-5 \times 12 \times -5) + (-7 \times -7 \times 8) + (18 \times -3 \times -3)]$$

$$= 664$$

$$\Delta_1 = [10 \times 12 \times 18 + (-3 \times -7 \times 0) + (-5 \times 0 \times -7)] - [(0 \times 12 \times -5) + (-7 \times -7 \times 10) + (18 \times 0 \times -3)]$$

$$\Delta_1 = 1670$$

$$I_1 = \frac{1670}{664} \approx 2.5 A$$

Subject:

موضوع الدرس:

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التاريخ: / /

$$I_3 = \frac{\begin{vmatrix} 8 & -3 & 10 \\ -3 & 12 & 0 \\ -3 & -7 & 0 \end{vmatrix}}{664}$$

$$\Delta_3 = [8 * 12 * 0 + (-3 * 0 * -3) + (10 * -3 * -7)] -$$

$$[(-5 * 12 * 10) + (-7 * 0 * 8) + (0 * -3 * -3)]$$

$$I_3 = 810$$

$$I_3 = \frac{810}{664} = 1.22 A$$

$$\therefore I \text{ at } 5 \Omega = I_1 - I_3 \Rightarrow 2.5 - 1.22$$

$$I \text{ at } 5 \Omega \simeq 1.28 A$$

Subject:

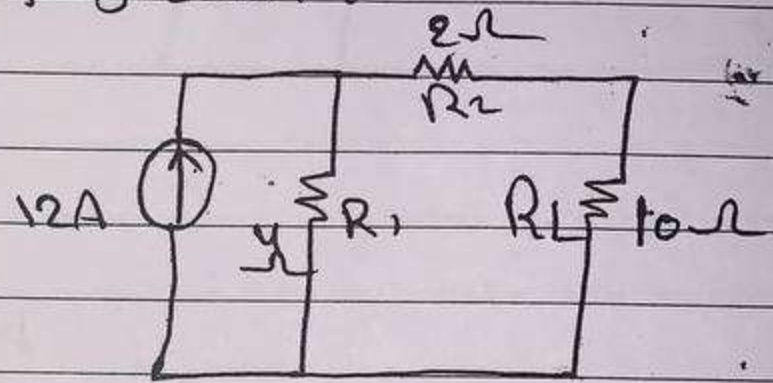
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موضوع الدرس:

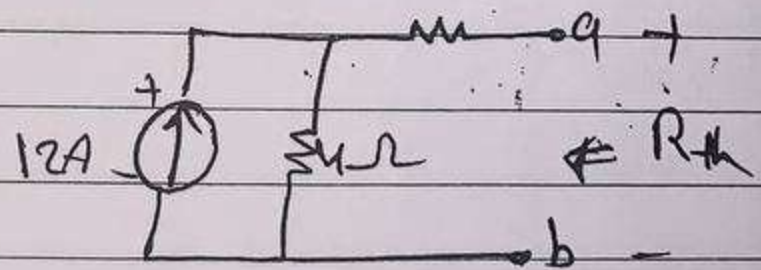
التاريخ: / /

Thevenins Theorem

Ex) Using thevenins theorem, find the current in the $R_L = 10\Omega$ of the network shown.



Step 1 & 2 → إزالة المقاومة المحددة، السؤال ووضع مكانها
أقطاب a و b



Step 3 → إزالة مصدر التيار وجدها في R_{th} المقامات القريبة فقط

$$R_{th} = R_1 + R_2 = 2 + 4 = 6\Omega$$

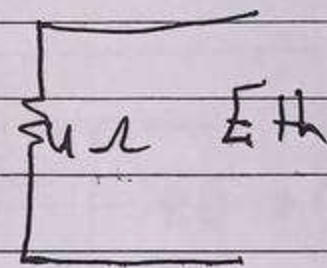
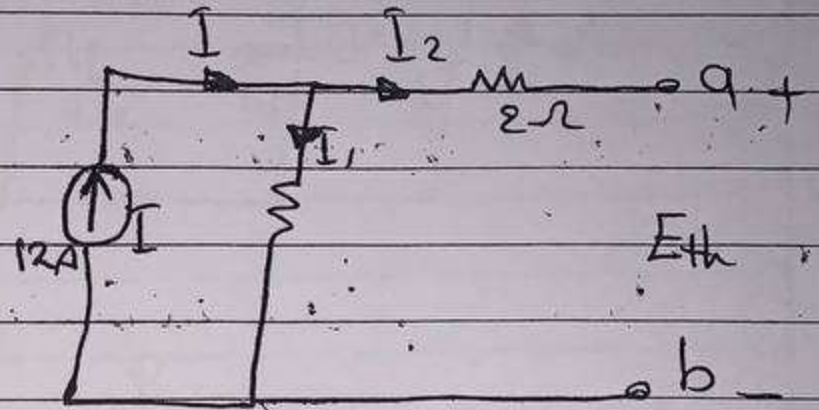
Step 4

ارجاعي المصادر الخارجيه الى صلي وبعدا
جد ال E_{th} حسب الرسم $I_2 = \text{open circuit}$

$$\therefore I_2 = 0A$$

$$V_2 = I_1 \times R_1$$

$$= 12 \times 4 = 48V$$

 $\therefore 2\Omega$ open circuit

$$\therefore 4\Omega // E_{th}$$

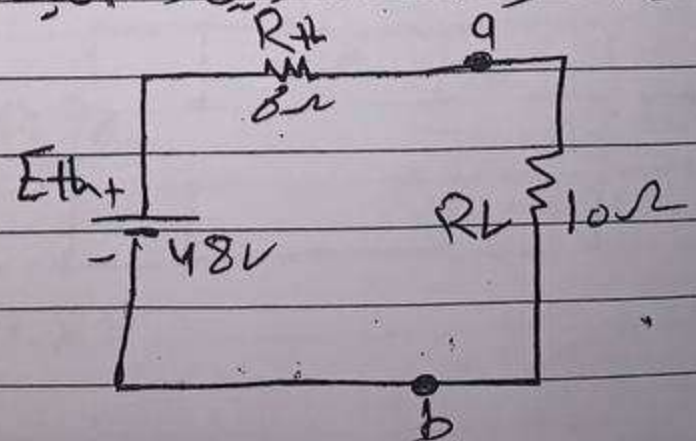
$$\therefore V_{R_2} = E_{th} = 48V$$

Step 5

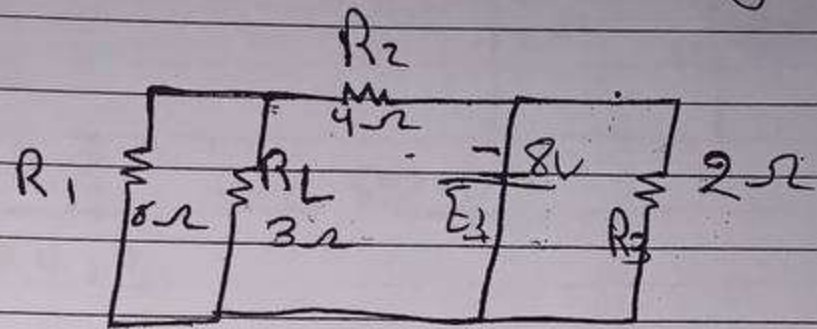
رسم الدارة المكافئة لإيجاد التيار

$$I = \frac{E_{th}}{R_L + R_{th}}$$

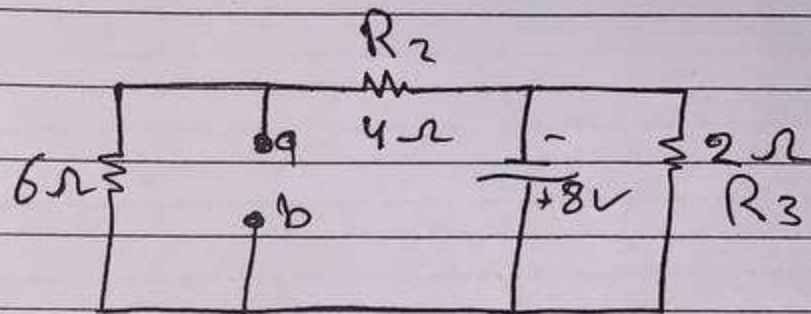
$$= \frac{48}{6 + 10} = 3A$$



Ex] find the current in the 3Ω resistor using thevenins theorem.

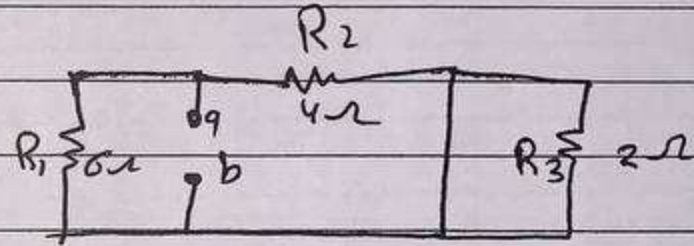


Step 1 and 2



Step 3: $R_{th} = ?$

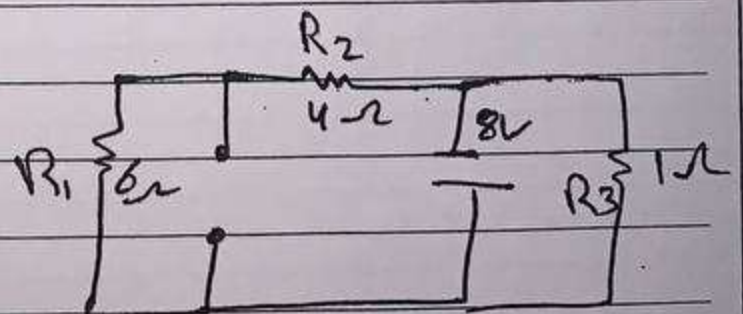
$$R_1 // R_2 = R_{th}$$



$$\frac{6 * 4}{6 + 4} = \frac{24}{10} = 2.4\Omega$$

Step 4: $E_{th} = ?$

using VDR



$$E_{th} = \frac{E_1 * R_1}{R_1 + R_2} = \frac{8 * 6}{6 + 4} = 4.8V$$

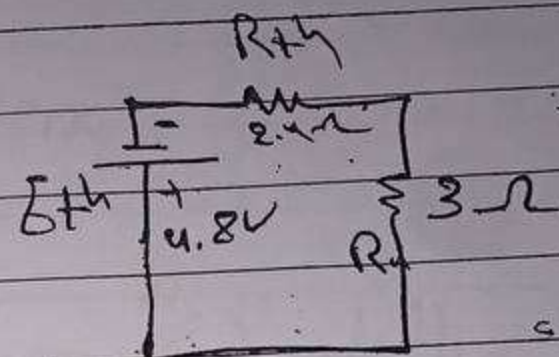
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التاريخ:

Step 5



$$I = \frac{E_{th}}{R_L + R_{th}} = \frac{4.8}{2.4 + 3} = 0.889A$$

Subject:

Date: / /

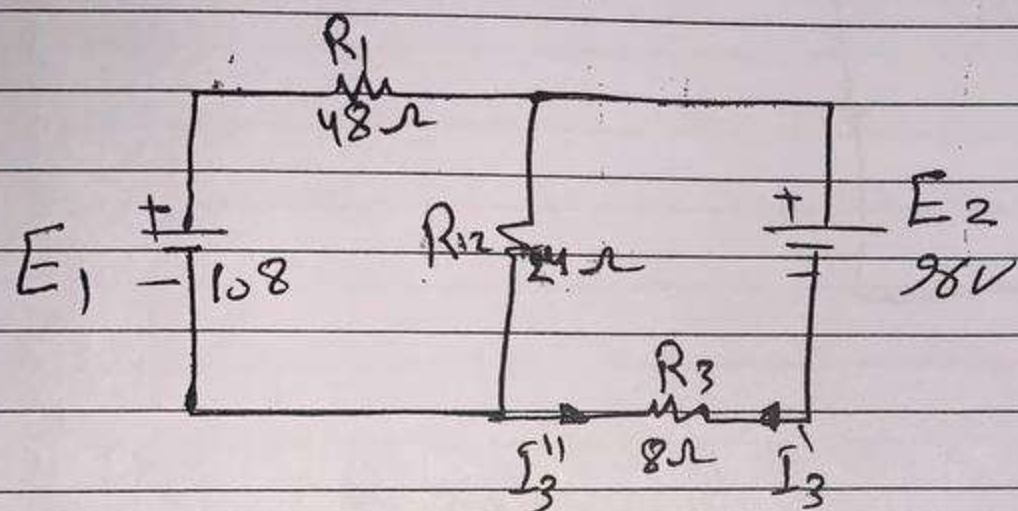
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موضوع الدرس:

التاريخ: / /

Superposition theorem

Ex) Using the superposition theorem ^{to} find the current through the 8Ω Resistor for the network shown.



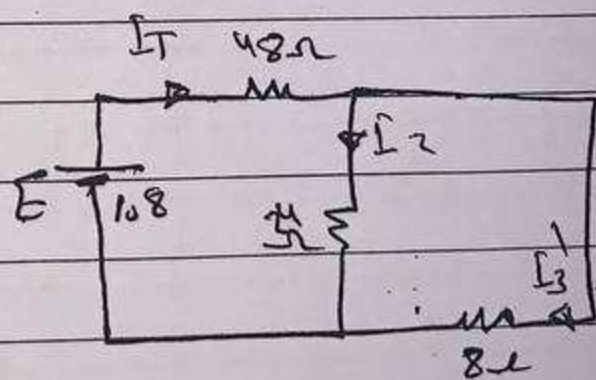
Sol) Consider the effect of E_1

$$R_T = (R_3 // R_2) + R_1$$

$$= (8 // 24) + 48$$

$$= 6 + 48 = 54 \Omega$$

$$I_T = \frac{E_1}{R_T} = \frac{108}{54} = 2A$$



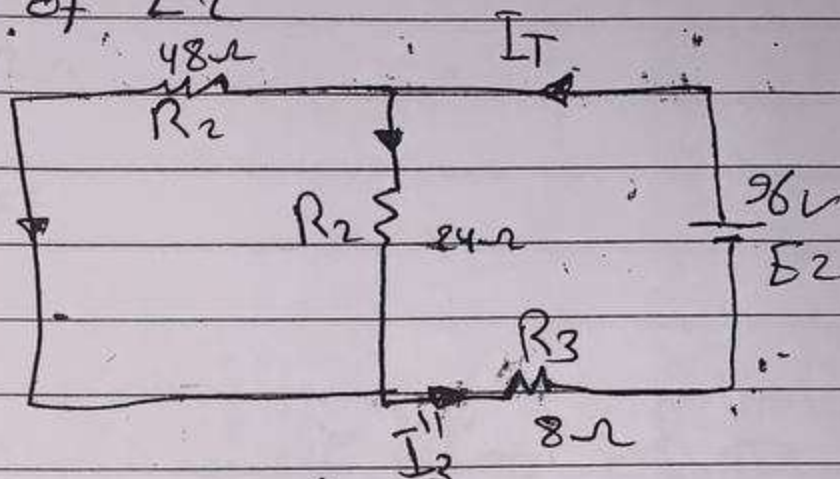
Subject:

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$$I_3 = \frac{I_T * R_2}{R_2 + R_3} = \frac{2 * 24}{24 + 8} = 1.5 A$$

Consider the effect of E_2 

$$R_T = (R_1 // R_2) + R_3$$

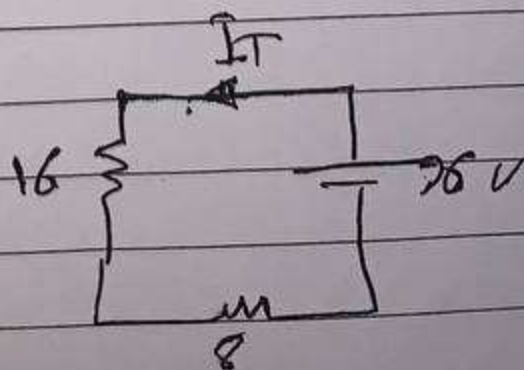
$$= (48 // 24) + 8$$

$$= 16 + 8$$

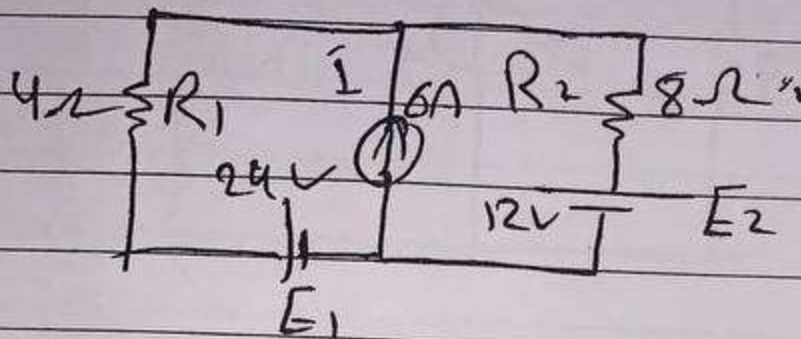
$$= 24 \Omega$$

$$I_T = \frac{E_2}{R_T} = \frac{26}{24} = 4 A = I_3''$$

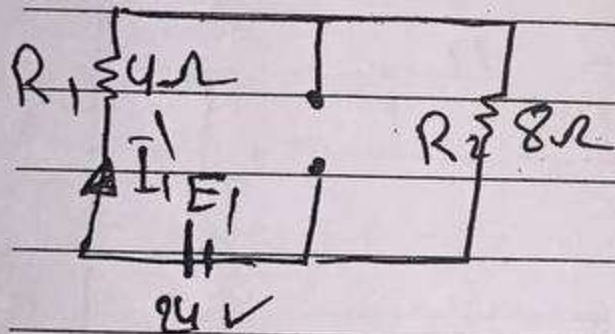
$$\therefore I_3 = I_3'' - I_3' = 4 - 1.5 = 2.5 A$$



Ex) Using the superposition theorem. find the current through the 4Ω resistor of the network shown



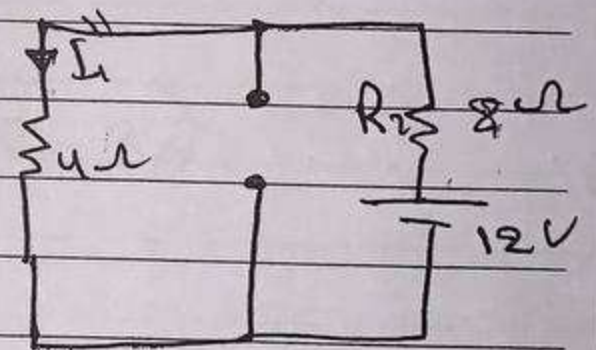
Effect of E_1



$$R_T = R_1 + R_2 = 4 + 8 = 12\Omega$$

$$I_1 = \frac{E_1}{R_T} = \frac{24}{12} = 2A$$

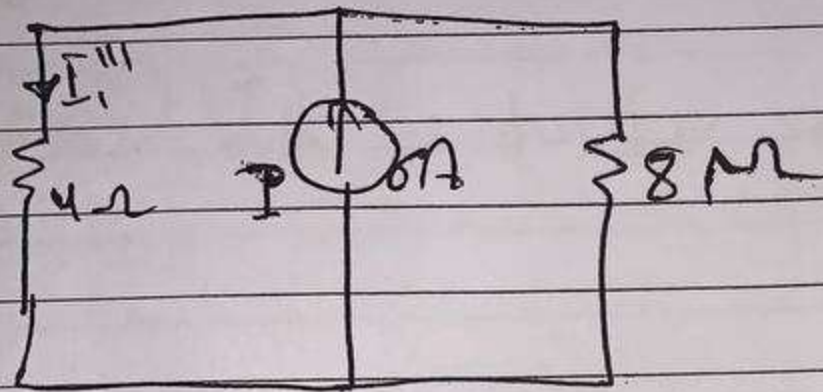
Effect of E_2



$$R_T = 4 + 8 = 12\Omega$$

$$I_T = \frac{E_2}{R_T} = \frac{12}{12} = 1A$$

The effect of I



$$I_1''' = \frac{I_T \times R_2}{R_2 + R_1} = \frac{6 \times 8}{8 + 4} = \frac{48}{12} = 4A$$

$$I_1 = I_1'' + I_1''' - I_1' = 1 + 4 - 2 = 3A$$

Subject:

موضوع الدرس:

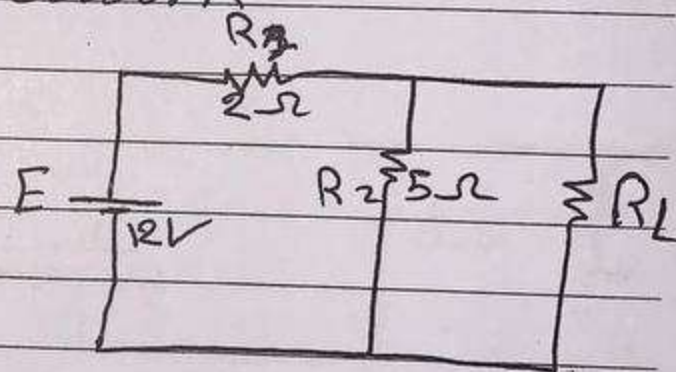
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التاريخ: / /

Norton theorem

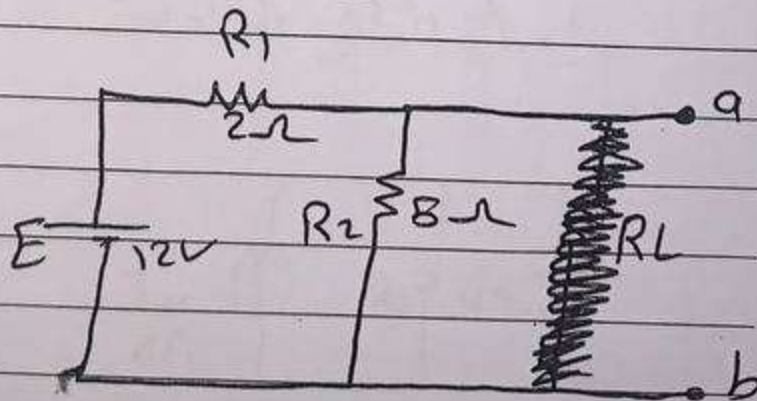
- هي تنسبة Thevenin ولكن نجد فيها R_N بدل من R_{th} ونجد E_N بدل E_{th} ونجد I_N بدل من V_{th} (E_{th})

Ex) for the circuit shown, find the Norton equivalent circuit for the network



Sol) step 1 and 2

إزالة المقاومة ومنع اقتران a و b



Subject: _____

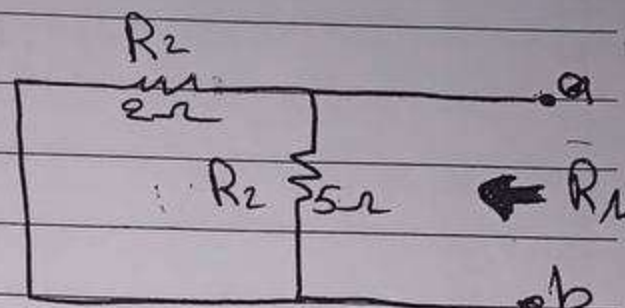
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التاريخ: / / _____

Step 3 → find R_N

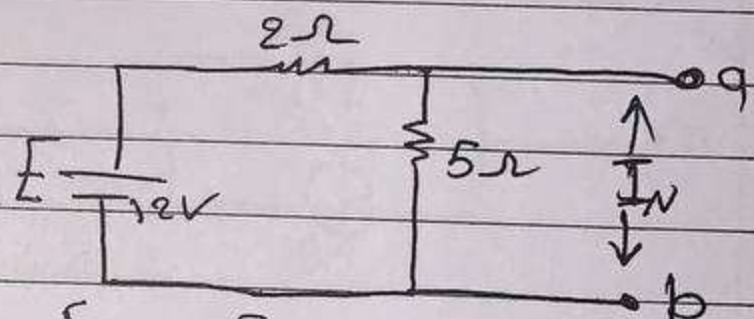
$$R_N = R_1 // R_2$$



$$\therefore R_N = \frac{1}{\frac{1}{2} + \frac{1}{5}} = \frac{2 \times 5}{2 + 5} = \frac{10}{7}$$

$$R_N = 1.43 \Omega$$

Step 4 → I_N



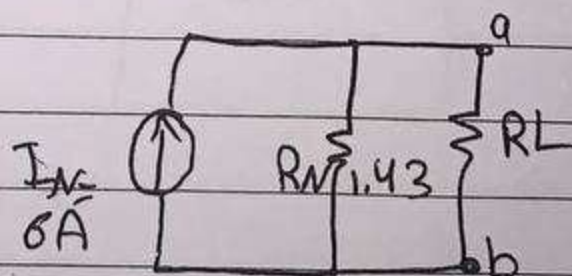
$$I_N = \frac{12}{2} = 6A$$

$$I_T = I_{5\Omega} = I_N$$

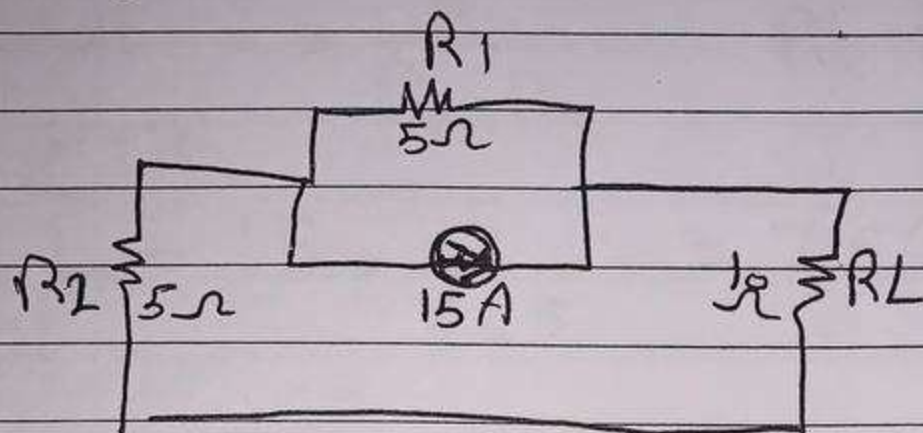
because $I_{5\Omega} = 0$ because open circuit

Step 5 →

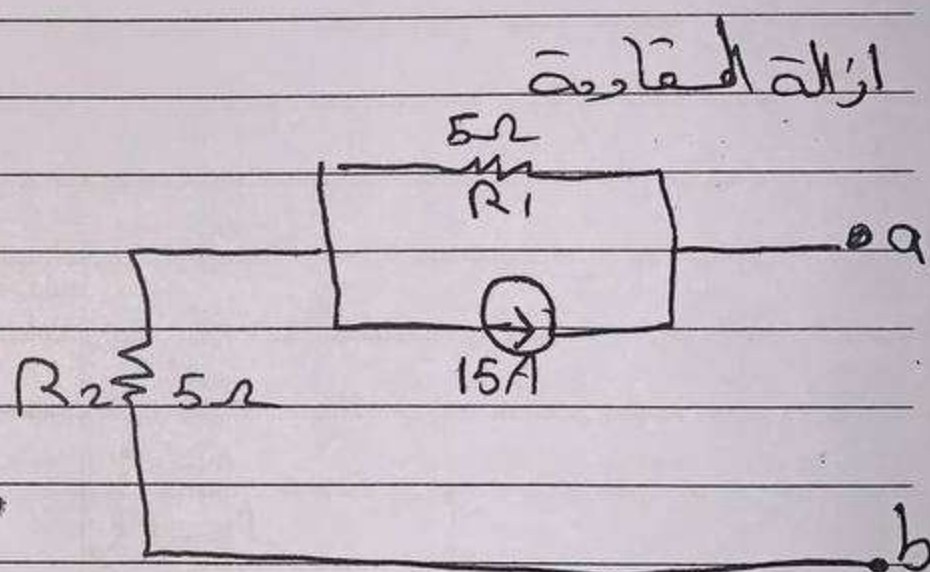
إيجاد الدائرة الكافئة



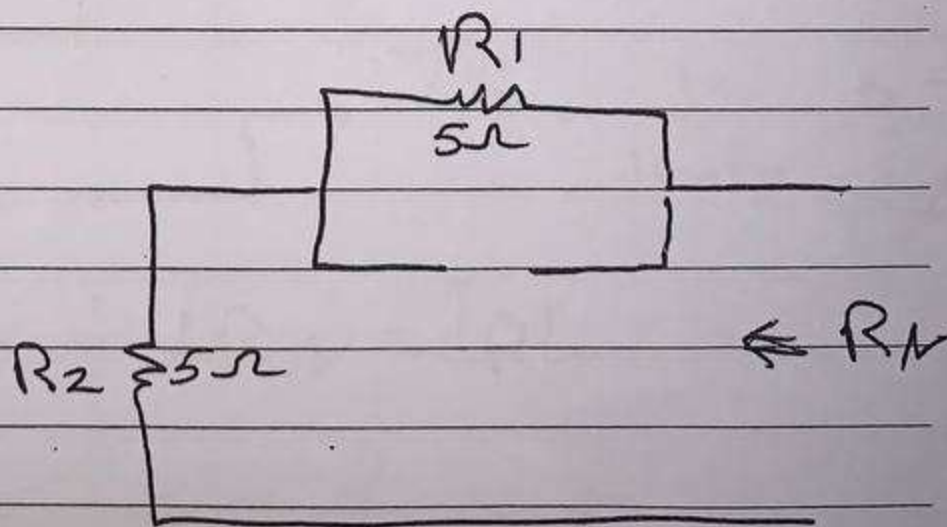
Ex) Using Norton theorem to find the current through the Load resistor R_L :



Step 1 and 2



Step 3: $R_N = ?$



$$R_N = R_1 + R_2$$

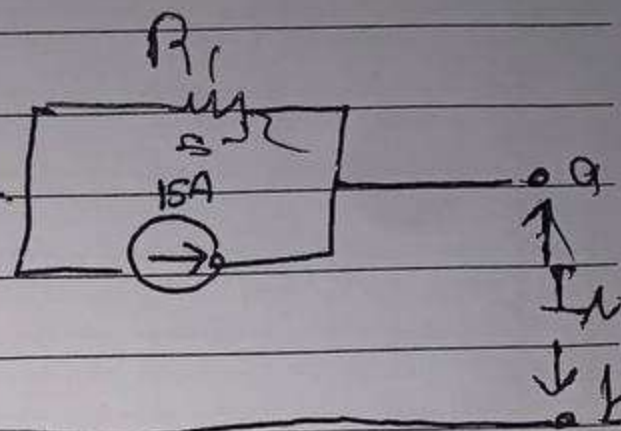
$$R_N = 5 + 5$$

$$R_N = 10\Omega$$

Step 4 $\Rightarrow I_N = ??$

$$I_N = I \cdot \frac{R}{R_1 + R_2}$$

$R_2 = 5\Omega$

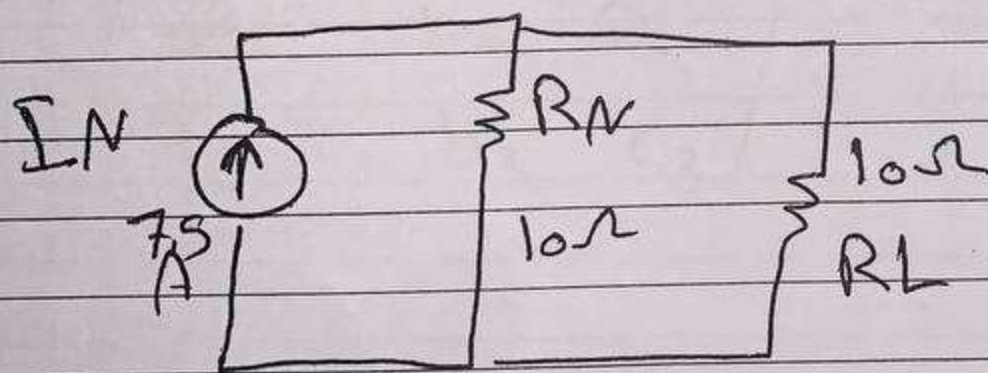


CDR

$$I_N = 15 \cdot \frac{5}{5+5} = 7.5A$$

$\therefore \textcircled{2} \therefore I_N = I_{R_2}$ because series

step 5



CDR find I_{RL}

$$7.5 \cdot \frac{10}{10+10} = 3.75$$

7.5

$$\therefore R_N = R_L = 10\Omega \therefore I_{R_N} = I_{R_L}$$

Subject:

Date: / / الموافق:

موضوع الدرس: التاريخ: / /

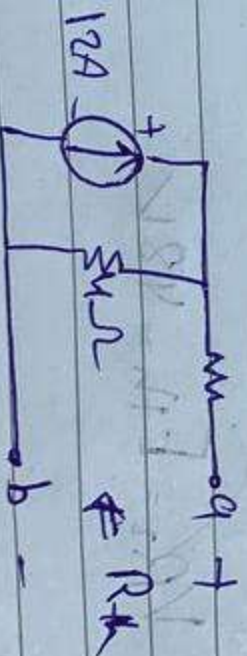
Theremin's Theorem

Ex) Using theremin's theorem, find the current in the $R_L = 10\Omega$ of the network shown.



Step 1 & 2 →

إزالة المقاومة المحملة، لسهولة الحساب
الخطاب هو a



Step 3 → إزالة المقاومة المحملة و حساب R_{th} من أجل التيار القوي

$$R_{th} = R_1 + R_2 = 2 + 4 = 6\Omega$$

Subject:

Date: / /

الموافق:

موضوع الدرس:

التاريخ: / /

Step 4

الخطوة الرابعة: إيجاد الجهد الكهربائي E_{th} في terminals a-b

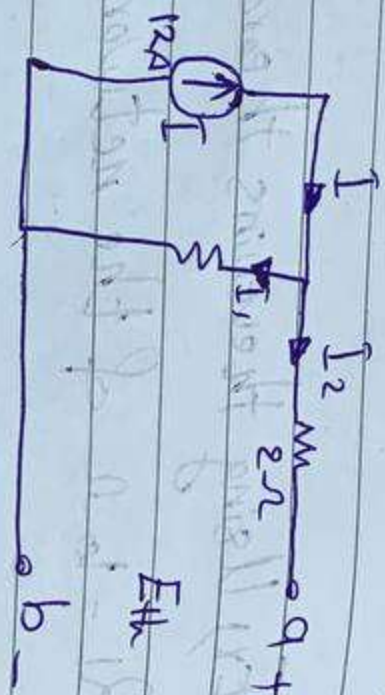
$I_2 = \text{open circuit}$

$\therefore I_2 = 0A$

$V_R = I_1 \times R_1$

$= 12 \times 4 = 48V$

$\therefore 2\Omega$ open circuit



$\therefore 4\Omega \parallel E_{th}$

$\therefore V_{R2} = E_{th} = 48V$

Step 5

الخطوة الخامسة: إيجاد التيار I في terminal a-b

$$I = \frac{E_{th}}{R_L + R_{th}}$$

$$= \frac{48}{4 + 10} = 3A$$

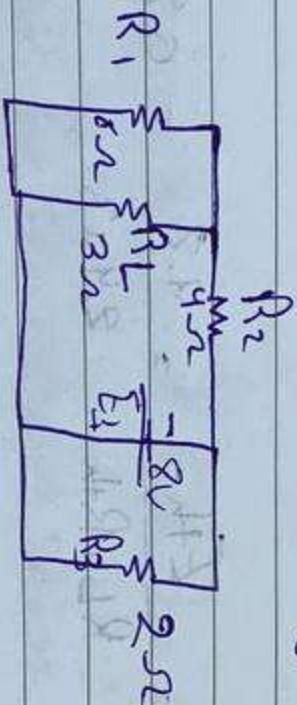


Subject:

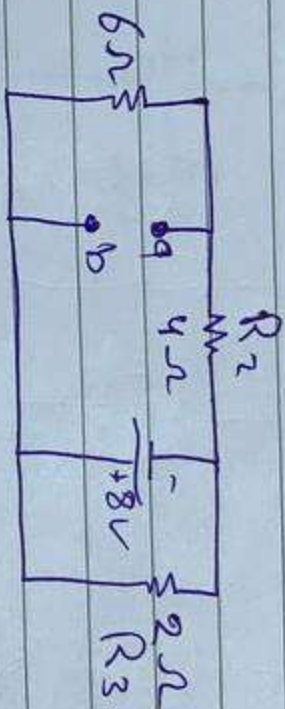
Date: / / الموافق:

موضوع الدرس: التاريخ: / /

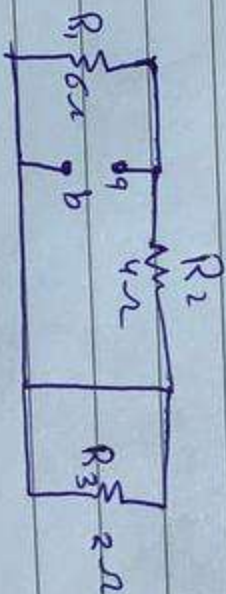
Ex1 find the current in the 3Ω resistor using thevenins theorem.



Step 1 and 2



Step 3: $R_{th} = ?$



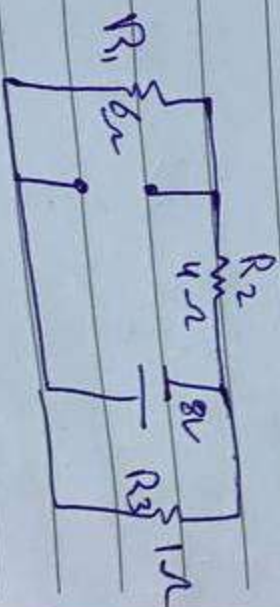
$$R_1 // R_2 = R_{th}$$

$$\frac{6 * 4}{6 + 4} = \frac{24}{10} = 2.4\Omega$$

Step 4: $E_{th} = ?$

using VDR

$$E_{th} = \frac{E_1 * R_1}{R_1 + R_2} = \frac{8 * 6}{6 + 4} = 4.8V$$



Subject: _____

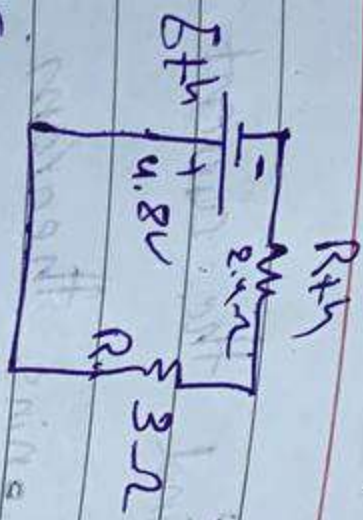
Date: / / الموافق: _____

موضوع الدرس: _____

التاريخ: _____

Step 5

$$I = \frac{E_{th}}{R_L + R_{th}} = \frac{4.8}{2.443} = 0.8829A$$



Subject:

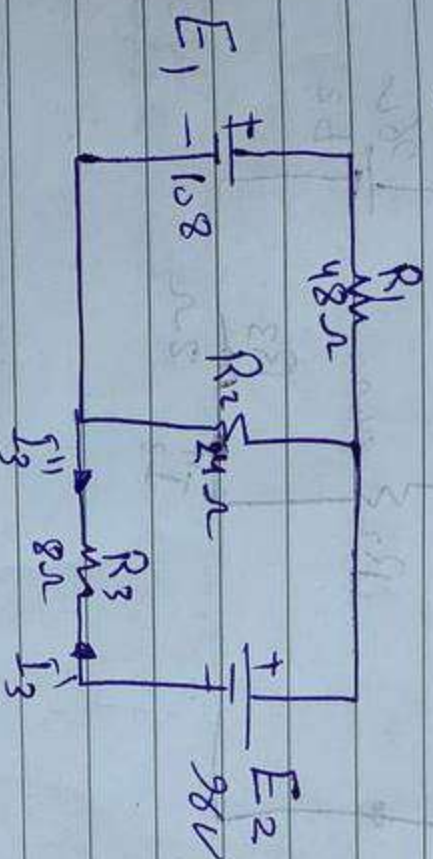
Date: / / : الموافق

موضوع الدرس:

التاريخ: / /

Superposition theorem

Ex) Using the superposition theorem ^{to} find the current through the 8Ω resistor for the network shown.



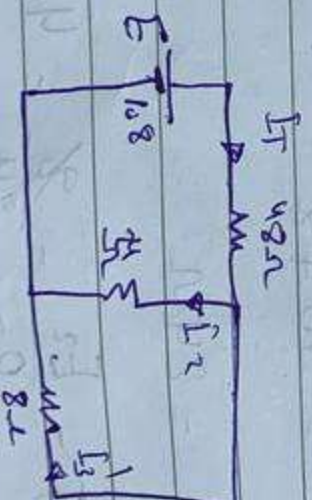
Sol) Consider the effect of E_1

$$R_T = (R_3 // R_2) + R_1$$

$$= (8 // 24) + 48$$

$$= 6 + 48 = 54 \Omega$$

$$I_T = \frac{E_1}{R_T} = \frac{108}{54} = 2A$$



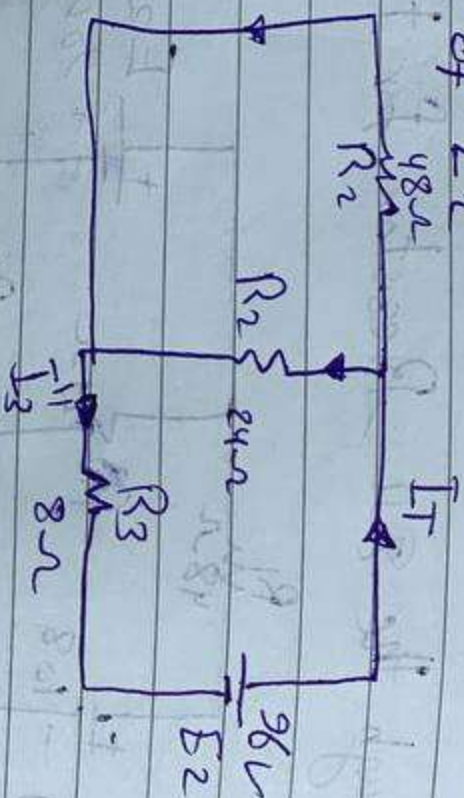
Subject: _____

Date: / / الموافق:

موضوع الدرس: التاريخ:

$$I_3 = \frac{I_T * R_2}{R_2 + R_3} = \frac{2 * 24}{24 + 8} = 1.5 A$$

Consider the effect of E_2



$$R_T = (R_1 // R_2) + R_3$$

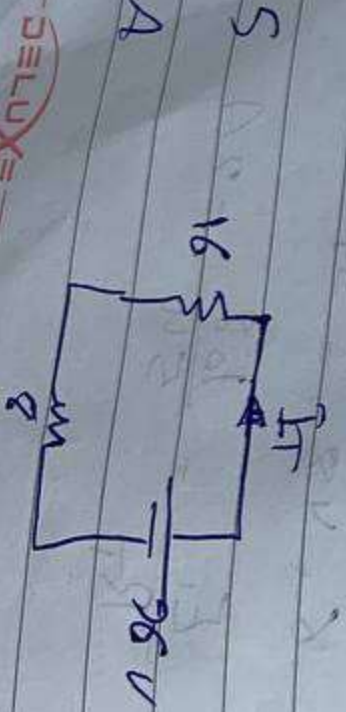
$$= (48 // 24) + 8$$

$$= 16 + 8$$

$$= 24 \Omega$$

$$I_T = \frac{E_2}{R_T} = \frac{26}{24} = 1.1 A = I_3$$

$$I_3 = I_3'' - I_3' = 1.1 - 1.5 = -0.4 A$$



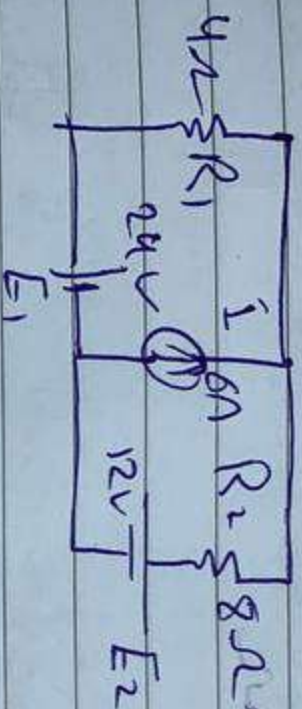
Subject:

موضوع الدرس:

Date: / / المواعيد:

التاريخ: / /

Ex) Using the superposition theorem. find the current through the 4Ω resistor of the network shown

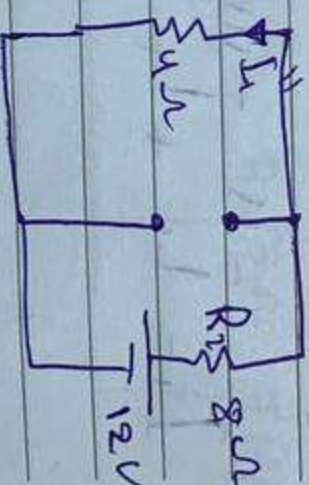
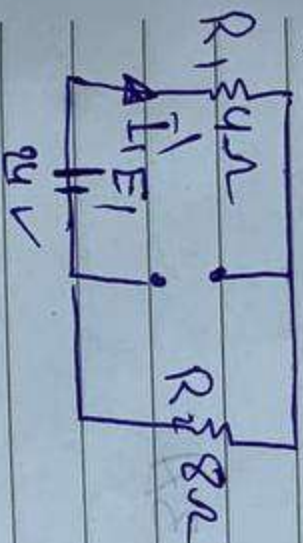


15V

Effect of E_1

8Ω
 $4 + 8$

Effect of E_2



$$R_T = R_1 + R_2 = 4 + 8 = 12\Omega$$

$$R_T = 4 + 8 = 12\Omega$$

$$I_1 = \frac{E_1}{R_T} = \frac{24}{12} = 2A$$

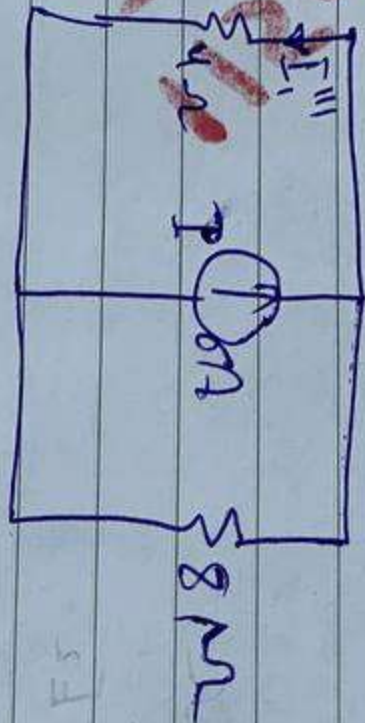
$$I_2 = \frac{E_2}{R_T} = \frac{12}{12} = 1A$$

Subject: _____

Date: / / الموافق:

موضوع الدرس: التاريخ:

The effect of I



$$I_1^{III} = \frac{I_T \times R_2}{R_2 + R_1} = \frac{6 \times 8}{8 + 4} = \frac{48}{12} = 4A$$

$$I_1 = I_1^{II} + I_1^{III} - I_1' = 1 + 4 - 2 = 3A$$

Subject:

Date: / / الموافق:

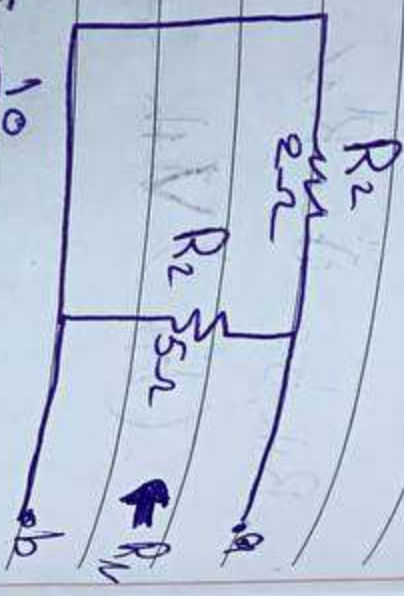
موضوع الدرس: التاريخ:

Step 3 → find R_N

$$R_N = R_1 // R_2$$

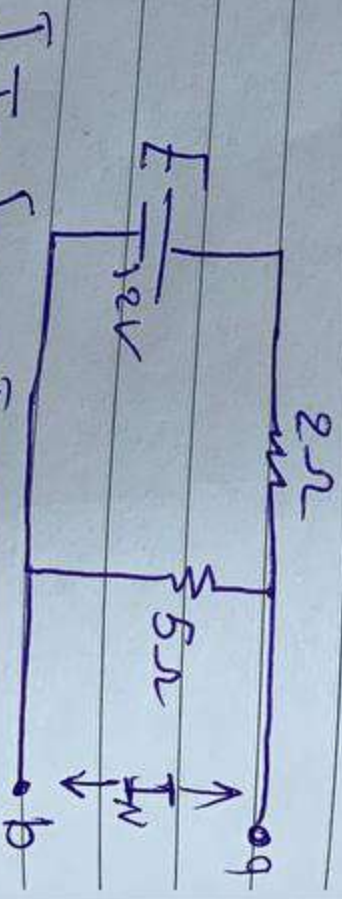
$$\therefore R_N = \frac{1}{\frac{1}{2} + \frac{1}{5}} = \frac{2 \times 5}{2 + 5} = \frac{10}{7}$$

$$R_N = 1.43 \Omega$$



Step 4 → I_N

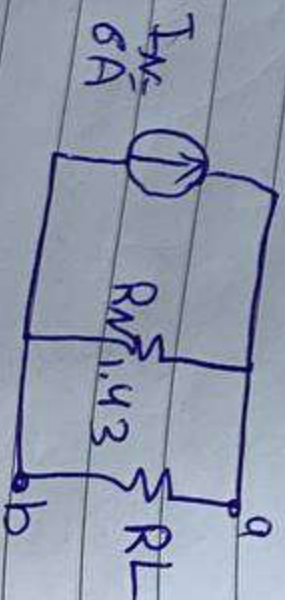
$$I_N = \frac{12}{2} = 6A$$



$I_T = I_{sc} = I_N$
because $I_{sc} = 0$ because open circuit

Step 5 →

إيجاد الدائرة المكافئة

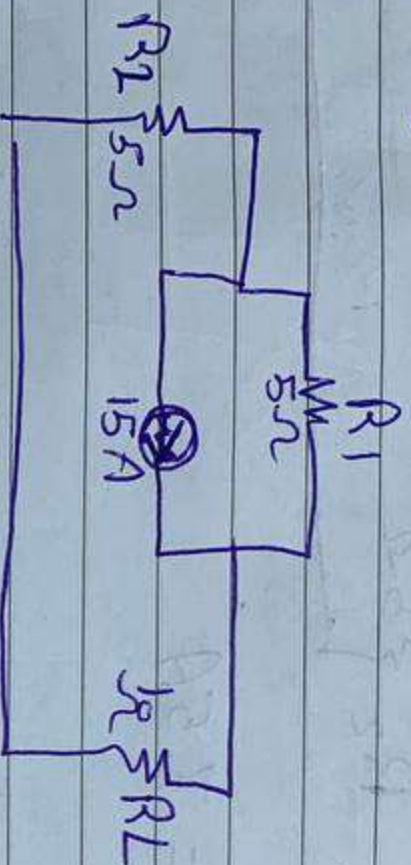


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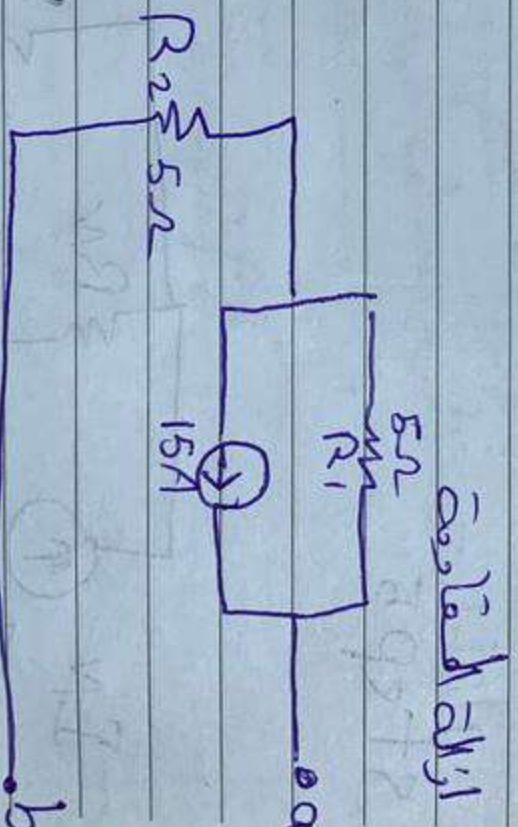
Date: / / الموافق:

موضوع الدرس: التاريخ: / /

Ex1 Using Norton theorem to find the current through the load resistor R_L :

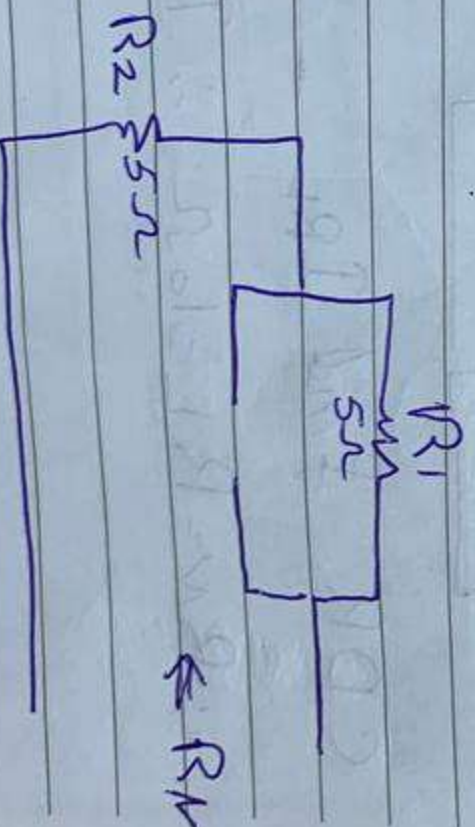


Step 1 and 2



إزالة المقاومة

Step 3: $R_N = ?$



$$R_N = R_1 + R_2$$

$$R_N = 5 + 5$$

$$R_N = 10\Omega$$

Subject:

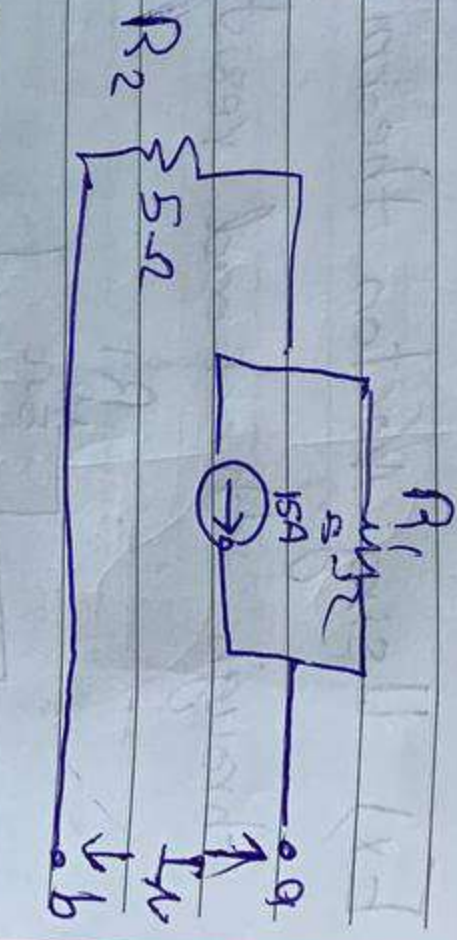
Date: / / الموافق:

موضوع الدرس:

التاريخ: / /

Step 4 $\Rightarrow I_N = ??$

$$I_N = I \cdot \frac{R}{R_1 + R_2}$$



$$I_{R_2} = I_N = 15 \cdot \frac{5}{5+5} = 7.5A$$

$\therefore I_N = I_{R_2}$ because Series

Steps



CDR find $I_{R_L} \rightarrow 7.5A$

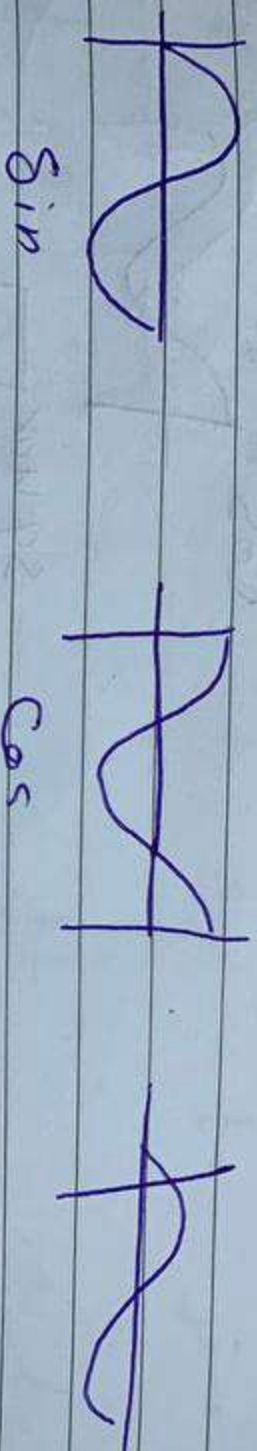
$$\therefore R_N = R_L = 10\Omega \therefore I_{R_N} = I_{R_L}$$

Subject:

Date: / / الموافق:

موضوع الدرس: / / التاريخ:

التيار البديل : Alternative Current



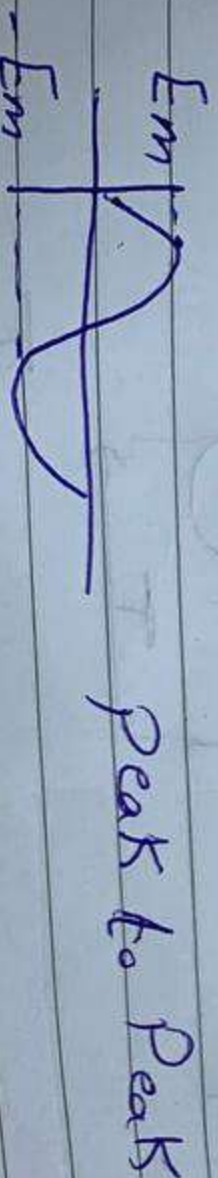
تيار فولتية غير مستقرة

Sin V soidal Voltage or current
التيارات الجيبية أو الفولتية الجيبية

$$E = E_m \sin \omega t$$

$$I = I_m \sin \omega t = I_m \sin 2\pi f t$$

Amplitude, E_m , I_m
هو مقدار الطاقة
التيارة لنقل الفولتية
أو التيار عن مكان إلى آخر



Subject: _____

Date: / / الموافق:

موضوع الدرس: التاريخ: / /

2] Waveform أشكال موجة

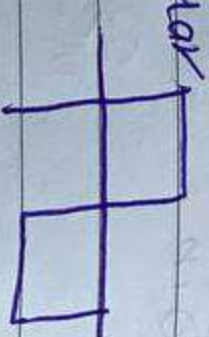
Sin



cos



Square



التردد الزاوي
③ ω : angular frequency = $2\pi f$

هو عدد المرات التي يغير فيها الجهد أو التيار 2π مرة

2π : Cycle = 360



or



$f = \frac{1}{T}$



$T = \text{one cycle}$

الموضوع: الدرس: التردد 50 Hz إذا قررنا أن نكتبه في شكل الفرق الجهد

Date: / / المواعيد: / / التاريخ: / /

التردد 50 Hz

موضوع الدرس: / / التاريخ: / /

الفرق الجهد $V = V_m \sin(\omega t + \phi)$ أو $V = V_m \sin(\omega t - \phi)$

التردد f : $\omega = 2\pi f$

الزاوية ϕ : Phase

الفرق الجهد $V = V_m \sin(\omega t + \phi)$

التردد f : $\omega = 2\pi f$

الزاوية ϕ : Phase

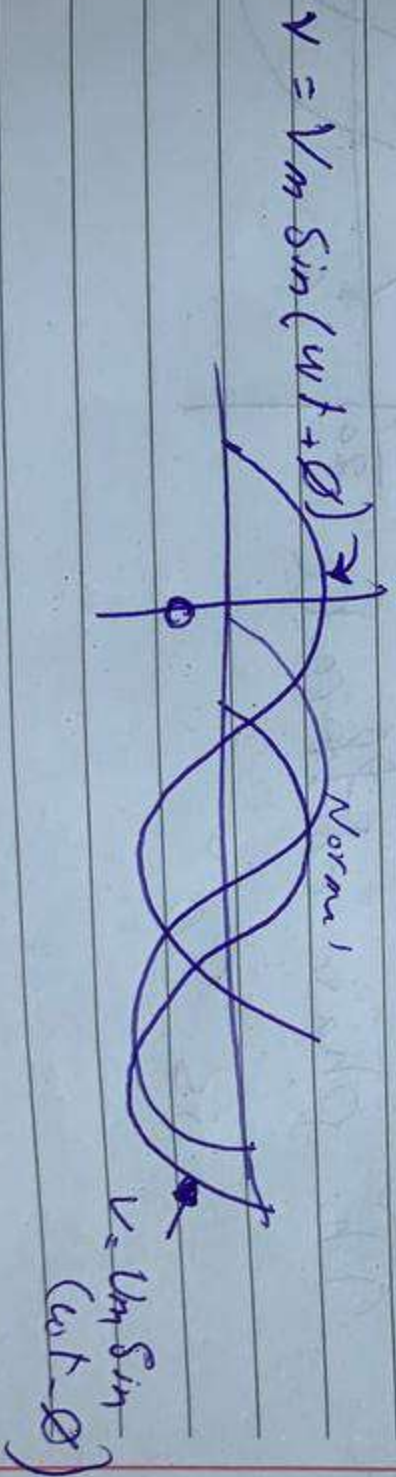
$$E = E_m \sin(\omega t + \phi)$$

$$E = E_m \sin(\omega t + 0)$$

$$E = E_m \sin(\omega t + 90^\circ)$$

$$E = E_m \sin(\omega t + 60^\circ)$$

$$E = E_m \sin(\omega t - 30^\circ)$$



Subject:

Date: / /

الموافق: 15/5/2022

الموضوع: الدرس: / / التاريخ: / /

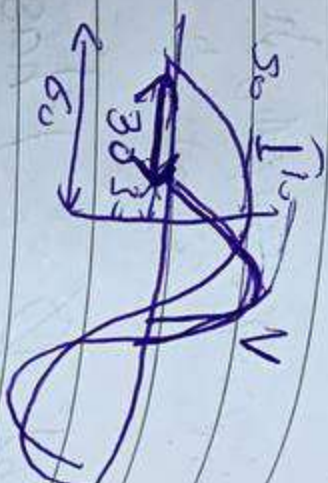
Phase difference

فارق الطور

ex

$$V = 10 \sin(\omega t + 30^\circ)$$

$$I = 5 \sin(\omega t + 60^\circ)$$



ex) The Phase difference = 30°

$$V = 8 \sin(\omega t - 60^\circ)$$

$$I = 8 \sin(\omega t + 20^\circ)$$

$$= 60^\circ - (-20^\circ)$$
$$= 80^\circ$$

The phase difference is 80°

